

Chapter 12

Bank risk management

Learning objectives

- To define risk measurement and risk management
- To understand the role of corporate governance in determining banks' risk culture
- To understand the importance of risk management
- To identify the main risk management techniques

12.1 Introduction

This chapter focuses on risk management as a central management tool to ensure banks' soundness and profitability. Risk management is a complex and comprehensive process, which includes creating an appropriate environment, maintaining an efficient **risk measurement** structure, monitoring and mitigating risk-taking activities and establishing an adequate framework of internal controls. As we have noted in previous chapters, the management of banking risks became increasingly important in the light of the 2007–2009 global financial crisis, that many considered the result of a failure of **risk management** systems. The need for more effective risk management and stronger internal controls in banking is being discussed by policy makers globally. In particular, one issue that has come to the spotlight relates to the need to tackle the problem of excessive bank risks at its roots, by setting standards of good practice in relation to **corporate governance** and regulating bank managers' remuneration packages. In addition, the Basel Committee on Banking Supervision (BCBS) has proposed a series of capital and liquidity reforms to make banks safer (see also Chapter 7). Among the key objectives are raising the quality, consistency and transparency of the capital base; enhancing banking risk coverage; introducing a leverage ratio; promoting forward-looking provisioning and holding buffers above the minimum required. Accordingly, banks will not only be required to adopt more formal and quantitative risk measurement and risk management procedures and processes but will also have to make sure that they adopt appropriate risk practices and be more transparent in relation to their risk exposure. Another related crucial aspect concerns the financial instruments used for risk management purposes,

whose notional value may not be directly related to the extent of bank risk exposure. We have discussed derivative products extensively in Chapter 10 and, although not discussed in this chapter, the reader should be aware of their key role in the banks' risk management process.

It is not only regulators that have placed an increased emphasis on risk management in an attempt to foster financial stability and economic development, it is also all the more important for bankers to manage their capital more efficiently in order to maximise risk-adjusted returns from their business activities. An emerging role in risk management is that of sustainability, as recent research has found that it is associated not only with lower default risk but also systemic risk (Scholtens and van't Kooster, 2019). In addition, society and regulators' increased expectations about ESG matters mean that long-term strategies will have to be integrated into banks' business models and risk management.

There are several aspects of risk management in banking and this chapter does not aim to analyse them all in detail, but rather to highlight the main issues. While Section 12.2 introduces the general issues of risk management, Sections 12.3 to 12.6 outline the main techniques used by banks to manage risks including credit risk, interest rate risk and liquidity risk. Sections 12.7 and 12.8 illustrate the techniques that are used to manage market and operational risk. Finally, the management of country risk is discussed in Section 12.9.

12.2 General risk management

This section focuses on how the risk management function is handled within the banking organisation and highlights the importance given to this function by managers and the institutional environment influencing its effectiveness and efficiency. Here we aim to provide an overview of systems and practices that cut across the major types of risks faced by banks. These systems and processes include such items as the allocation of resources to risk management activities, governance issues, record-keeping, communications within the organisation and internal audit. As pointed out by Cumming and Hirtle (2001) the difference between risk measurement and risk management is that while risk measurement deals with the quantification of risk exposures, risk management deals with the overall process that a financial institution follows to define a business strategy, to identify the risks to which it is exposed, to quantify those risks and to understand and control the nature of risks. In Chapter 11 we reviewed the main risks faced by financial institutions; for each class of risk, banks need to estimate the expected losses and the probability of unexpected losses, so that an appropriate amount of capital may be held.

It is important to recall that the main objective of bank risk managers is that of shareholders' wealth maximisation and in pursuit of such an objective they have to manage carefully the trade-off between risk and returns. In order to increase shareholders' wealth a company has to generate returns greater than its opportunity cost of capital. The opportunity cost of capital is the perceived cost to the bank of raising equity and keeping shareholders happy (see also Section 9.4.3). For example, if a bank makes an acquisition that generates a ROE of 8 per cent, but the cost of obtaining the capital funds needed to undertake the acquisition is 10 per cent then this destroys shareholders' wealth. Alternatively, if returns exceed 10 per cent then the acquisition creates value for shareholders. Typically, higher returns (ROE) are reflected in higher market valuations of a company's shares – or to put it another way, investors rank profitable firms more highly and this is reflected in greater equity prices. The aim of bank managers therefore is to maximise ROE relative to its cost of capital and this will maximise

shareholders' wealth. Note that banks can do these sorts of calculations for all parts of their business in order to identify how capital is being used within the bank and what parts are generating the best or worst returns.

Investors, meanwhile, can expect a higher rate of return only by increasing the risk they are prepared to bear. Risk measures are therefore related to profitability measures, as banks must take risks to earn adequate returns. As Sinkey (2002) points out, the essence of modern banking is the measuring, managing and accepting of risk and the heart of bank financial management is risk management. The task then becomes: how to set appropriate targets for a bank's returns and the corresponding risks undertaken? Hempel and Simonson (2008), while cautioning that there is no exact answer, suggest three steps:

- assess how other similar individual banks and groups of banks have made their risk/return decisions;
- compare the bank's performance measure to those of similar banks;
- set reasonable objectives against the backdrop of a bank's historic performance, the performance of its peers and its external environment.

These steps, in turn, are essentially based on the following analysis:

- stock market expectations (if the bank is quoted);
- trend analysis of past performance;
- trend and comparative analysis of peers' performance allowing for factors such as business mix, available production technology and external environment (macroeconomic and regulatory).

One of the underlying issues in most banking systems is the fact that deregulation, globalisation and internationalisation have increased the (real or perceived) degree of competition in banking markets, requiring banks to take on more risk to achieve satisfactory returns.

Given their special role in any economy, banks should be run in a safe and sound manner. As discussed in Section 8.3.2, US regulatory authorities monitor banks' behaviour and try to ensure that they achieve a good CAMELS rating (adequate Capital, good Asset quality, competent Management, good Earnings, sufficient Liquidity and Sensitivity to market risk). Banks are rated 1 (essentially sound) to 5 (basically insolvent). Banks with ratings of 1 or 2 are considered to present few, if any, supervisory concerns, while banks with ratings of 3, 4 or 5 present moderate to extreme degrees of supervisory concern. A bank's CAMELS rating is highly confidential; it is evaluated by the bank's senior management and disclosed only to the appropriate supervisory staff. However, the public may infer such information based on subsequent bank actions or specific disclosures.

The main elements of modern risk management processes and strategies include identifying, measuring and monitoring risk exposures. The overall risk management process should be a comprehensive one, which creates a risk management culture in all departments of the financial institution. The specific asset-liability management (ALM) function (its coverage and functions) varies from bank to bank. However, as detailed in Chapter 10, the ALM function takes an overall risk management view of the bank. Specifically, ALM is concerned with explicit managerial and risk functions such as liquidity, capital management, funding and cost of funds, and managing the bank's security portfolio.

Interest rate risk management and lending and credit risk management are key components of a bank's overall risk management function, and are also part of the ALM function, but they are usually managed specifically by separate units or divisions within a bank. For

this reason, and because of the relative importance of such risks for financial institutions, we will analyse in detail their management processes. Before we move on to specific risk management techniques, let us outline the basic concepts of the risk management process.

Following the publication of the comprehensive set of ‘core principles’ for effective banking supervision in 1997 (BCBS, 1997a) and as part of an ongoing effort to enhance sound practices in banking organisations, the BCBS has issued a number of publications – in the form of core principles, standards and guidelines – to highlight the practices and criteria that should underpin the risk management process. The BCBS’s response to the 2007–2009 crisis has resulted in the compilation of a set of documents to build more resilient banks and a safer financial system that form the basics of a new Accord (Basel III). In 2017 a document was published that sets out the Basel Committee’s finalisation of Basel III (BCBS, 2017). Due to the Covid-19 pandemic, banks and supervisors have been granted a one-year deferral so these reforms will take effect from 1 January 2023 and will be phased in over five years. All global standards, including this one, and bank regulation and supervision, have been integrated in a new consolidated ‘**Basel Framework**’ that links them in one place.¹

It is noticeable that one of the most recent objectives of the Basel Committee is to evaluate the potential implications on individual banks’ safety and soundness and systemic financial stability derived from emerging risks in the area of sustainable finance (see Chapter 11). Key findings from a recent report suggest that climate-related financial risk should be addressed in existing regulatory and supervisory frameworks. Publications in these areas include:

- ‘Climate-related financial risks: A survey on current initiatives’ (BCBS, 2020a);
- ‘The green swan: Central banking and financial stability in the age of climate change’ (Bolton *et al.*, 2020).

As environmental challenges pose a potential strategic risk to banks, there is no doubt that corporate boards have a key role in the identification and management of these risk, as they do for other risks. Stroh (2019, 2020) identifies several challenges in this context related to time horizon, data gaps and complexity, as the impact of climate change is uncertain, non-linear and hard to predict. The European Banking Authority (2019) highlights how the governance of both public and private institutions play a crucial role in considering social and environmental issues. This includes internal management structures, employee relations and executive pay.

More generally, corporate governance and remuneration policies are particularly sensitive topics in banking for two main reasons: first because the **risk culture** shared in any corporation is ultimately determined and monitored by corporate governance frameworks. Second, because performance-related managers’ remuneration can potentially increase the risk-taking incentives and this can produce undesirable outcomes, particularly in a sector like banking that relies heavily on confidence. Box 12.1 looks into these issues in some detail; however, first it is useful for the reader to reflect on the concept of risk culture as defined in Table 12.1.

According to Power *et al.* (2016) the common thread that runs through almost all definitions of risk culture is that it concerns people’s behaviour within an organisation in relation to risk management. Corporate governance and leadership should design, express and impart the company’s propensity to risk. However, the governance implications in banks are different from other financial organisations for a number of reasons, as discussed in Box 12.1.

¹ The Basel framework is regularly updated (www.bis.org/basel_framework/). A full list of reports can be found at the Bank for International Settlements online publications archive at www.bis.org/bcbs.

Table 12.1 Selected definitions of risk culture

Definitions from the 'practice literature'	Source
A bank's norms, attitudes and behaviours related to risk awareness, risk-taking and risk management, and controls that shape decisions on risks. Risk culture influences the decisions of management and employees during the day to day activities and has an impact on the risks they assume.	Financial Stability Board (2014)
The values, beliefs, knowledge and understanding about risk shared by a group of people with a common purpose, in particular the employees of an organisation or of teams or groups within an organisation.	IRM (2012)
The combined set of individual and corporate values, attitudes, competencies and behaviour that determine a firm's commitment to and style of operational risk management.	BCBS (2011d)
The system of values and behaviour present throughout an organisation that shape risk decisions. Risk culture influences the decisions of management and employees, even if they are not consciously weighing risks and benefits.	KPMG (2010)
The norms and traditions of behaviour of individuals and of groups within an organisation that determine the way in which they identify, understand, discuss, and act on the risks the organisation confronts and the risks it takes.	IIF (2009)
Organisational behaviour and processes that enable the identification, assessment and management of risks relative to objectives ranging from compliance to operational, financial and strategic.	PwC (2009a)

Source: Adapted from Power *et al.* (2016).

BOX 12.1 THE CORPORATE GOVERNANCE OF BANKS

Since the financial crisis of 2007–2009, much has been said about the way in which banks are governed and how senior executives are remunerated. There is a general belief that some CEOs and other senior executives had perverse incentives that encouraged them to formulate strategies to pursue excessive risks (so as to boost their pay and other benefits) at the expense of the ultimate bank owners – the shareholders. There has also been criticism that some banks had too powerful CEOs that had undue influence on chairmen and boards resulting in little checks to their (sometimes) excessive risk-taking behaviour. Recent EU policy has been directed to limiting bank executive pay – by restricting bonus payments – and over the last decade or so there has been a general trend to improve the way in which publicly listed companies are governed. This section briefly highlights these topical issues relating to bank corporate governance.

How to define corporate governance?

There is no generally accepted definition of what corporate governance is, but typically it relates to the way outside investors and other stakeholders, such as employees and government, exercise control over

senior management and other corporate insiders in order to protect their interests. The main focus of the academic literature, prompted by the seminal work of Jensen and Meckling (1976), has tended to focus on the tensions that arise between corporate insiders and shareholders and also between equity holders and other creditors (such as banks and bondholders) over the appropriate amount of risk that firms (including banks) should take. As shareholders own the residual claims in the firms they invest in (this is a formal way of saying that creditors, like banks and bondholders, are paid before shareholders when firms make any payout – in both the good and bad times), so this structure creates a strong incentive for shareholders to emphasise risk-taking – shareholders want firms to take on risks to earn higher returns so that other creditors are first paid off and then they can earn higher dividends. More risk and greater returns will also likely boost share prices, thus feeding capital gains for shareholders. Of course, excessive risk-taking can lead ultimately to bankruptcy and failure – an outcome that shareholders and other stakeholders are unlikely to be keen on. The key issue is how to balance insider and outsider interests so firms do not



BOX 12.1 The corporate governance of banks (continued)

take on too much risk and the benefits of all parties are optimised.

Is the governance of banks different from other firms?

The situation of governance in banking is more complicated than for other types of firms for three main reasons:

- 1 Banks are highly leveraged – they have much less capital than other industries.
- 2 Banks have opaque business models – some activities are particularly difficult to evaluate, such as the amount of non-performing loans, the value of derivatives and trading activities, and so on.
- 3 Safety net subsidies, such as deposit insurance, too-big-to-fail and lender-of-last-resort backstops, mean that senior executives of banks may be incentivised to take on excessive risk as the downside is not too bad/limited. To put it another way, if senior executives take on too much risk and the bank fails they may not be too concerned as they believe that most depositors will be protected by deposit insurance and/or the government will bailout the bank if it's too-big-to-fail. So the upside of taking more risk (big bonuses and higher pay for executives) and the limited downside resulting from safety net subsidies, creates skewed incentives – encouraging bankers to take on too much risk.

Despite these differences, and as pointed out by Hagendorff (2019), traditionally, the corporate governance codes for banks were not much different from those applied to non-banking firms.

What are the functions of the board of directors?

The board of directors of publicly listed firms is regarded as the main internal control mechanism that promotes and protects shareholder interests. Typically, the main role of the board should be to act as a check on managerial excess, to monitor senior manager behaviour and to support management if the board sees fit. Boards also evaluate the performance of senior executives, decide on remuneration packages and

can fire poor performers. It is generally accepted that a board that has more outsiders (non-executive directors) with diverse backgrounds has the best chance of being independent in its actions. Typically, the more the board is insider controlled (for instance, if there is a high proportion of company senior managers on the board), the more likely it is to be captured by incumbents and the more likely it is to act on behalf of insiders (senior company managers). In contrast, the more independent the board, the more likely it is that the company will be run in the interests of shareholders. (There is a big consulting industry that advises Chairmen and CEOs on the make-up of their boards).

What are the key changes in respect of corporate governance since the banking crisis?

In February 2009, Sir David Walker was asked by the UK government to review the corporate governance of the UK Banking Industry (Walker, 2009). The report (which became known as the *Walker Review* and was published in November 2009) outlined the following key issues: role and constitution of the board; board size, composition and qualifications of board members; functioning of the board and evaluation of its performance; role of institutional shareholders; and the governance of risk and remuneration. Many of the recommendations emanating from the report have since been implemented – particularly those on pay and the qualifications of board members. Reviews of bank corporate governance have been carried out also in other countries. For example, in the Netherlands in 2010, new governance codes have been introduced to cover banks to complement those that apply to firms. A comprehensive survey of corporate governance of banks is provided by de Haan and Vlahu (2016).

Specifically on executive compensation, in 2010 the former FSA (now Prudential Regulation Authority) circulated the first version of a remuneration code that controls both level and structure of executive pay in all UK financial services firms (around 2,700 banks, building societies and some investment firms) (Financial Services Authority, 2010). The revised version of this code was issued in 2011 (Financial Services Authority, 2011b); however, although the UK is not part of the eurozone, as an EU member state it was subject to the CRD IV, the Fourth Capital Requirement Directive,



BOX 12.1 The corporate governance of banks (continued)

agreed in February 2013 by the European Parliament. This controversial legislative act was implemented in January 2014 and introduced a pay cap on bankers' bonuses throughout the EU. This is impacting a big chunk of the industry and has implications for centres that seek to attract highly paid financial sector employees, like the City of London. It should be noted that the decision to limit top bankers' pay has not been imitated in the US thus could lead to either movement of activity to the detriment of the City or to regulatory avoidance

via an increase in fixed pay 'allowances' on top of salaries for top bankers, with both treated as fixed pay. With Brexit, it became possible for the cap to be scrapped in the UK. However, since allowances are fixed pay (rather than variable like annual bonuses), they may have become attractive to staff, especially in light of the current economic uncertainty. Related to bankers' pay, an important issue concerns inequality and in particular the magnitude of gender pay gaps in the finance industry, as shown in the *FT* article reproduced below.

Trading floor culture remains a barrier for senior female staff (Article from the *FT*)

After her best year trading currencies, a female trader on a large US investment bank's foreign exchange desk was expecting good news at the bonus meeting. It was, after all, the late 1990s, the heyday of currency trading. But when her boss told her the sum, which was a fraction of what male members of the team were awarded, she stood up and went back to her desk in silence. When she later told her boss she was

disappointed, he turned around and snarled: "And how much does a nurse get?!" Incidents such as this have hampered a growing push to get more women into senior trading roles. In June this year, Mark Carney, governor of the Bank of England, said "finance offers so much, but it still doesn't offer enough for women", calling the lack of female representation in senior roles "an enormous missed opportunity". Andrew Hauser, the BoE's executive director for markets, said "diversity is not an option in

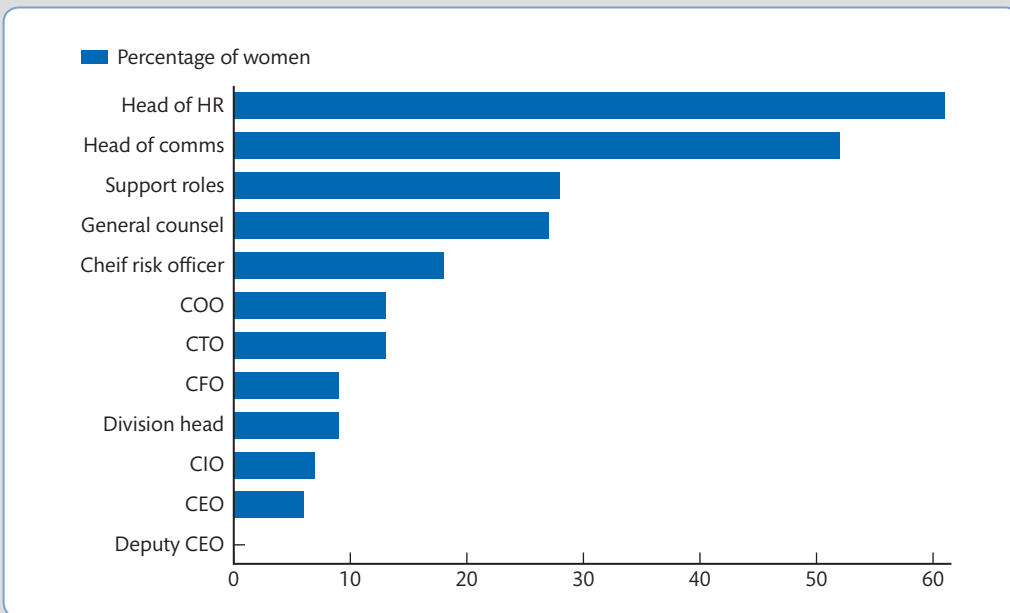


Figure 12.1 Gender breakdown of executive functions in UK financial services

Source: Szalay (2019).



BOX 12.1 The corporate governance of banks (*continued*)

FX — it's an imperative". Despite those efforts, only 13 per cent of individuals approved by the UK's Financial Conduct Authority for trading roles last year were female. Women foreign exchange traders have always been rare. Many have been put off by an aggressive and testosterone-fuelled atmosphere on the trading floor. As a result, few can boast careers that span decades. In trading and broking, fewer than 1 per cent of staff were historically female, according to an industry veteran.

The dangers of the overwhelmingly male culture in trading rooms have been exposed by the global probe into improper practices in currency markets, which focused on chat rooms with revealing names such as "Semi Grumpy Old Men". None of the chat rooms had female members and not a single woman faced criminal charges for the alleged failings. "Perhaps the most depressing aspect of past FX abuse was the light it threw on historic behaviour in parts of the market, called a 'bro-culture' by some," said Mr Hauser. An IMF study last year suggested that having more female traders "would foster greater stability in the banking system and enhance economic growth". It called the fewer than 2 per cent female representation in chief executive positions "shocking". A paper by the University of Leicester suggests that female traders outperform males on average, and that more women on the trading floor would reduce crash events. Companies say recruiting women is tough. In the UK last year fewer than 12 per cent of A-level candidates studying science, technology, engineering and mathematics were female. This is a huge restriction on the pool of talent that electronic trading groups look for, said Jessica Sohl, chair of Chicago-based computer trading specialist HC Technologies, where fewer than 5 per cent of employees are female. There is also a "legacy problem in retaining female talent in trading roles", said Simon Manwaring, global head of foreign exchange at NatWest Markets, who added that the industry is facing a supply-side

problem. "If I want to hire a senior options trader I would like them to have 15 years of experience. There are simply not many women around who have that," he pointed out. The culture in the 1990s was "pretty horrible" and "rife with sexism", said Kevin Rodgers, a retired former Deutsche Bank currencies head, who noted that women "quite sensibly stayed away". Those who went into trading often left after having a child, because the pressure of "having the seat empty" meant that six weeks' maternity leave was seen as a luxury. One trader said that while working in Oslo he sat next to a "brilliant" female peer for years, until one day she stood up and said that she was going on maternity leave. "We couldn't believe it. She kept it a secret, wearing big jumpers because she was worried about her job," he said. She never came back.

Worries over pay have not helped. Mr Manwaring said he was confident that in his business everyone was paid equally for equivalent jobs. This has not always been the case in the industry. The anonymous female trader said that she used to sit next to a man who was paid multiples of her salary for the same role, which made him feel so guilty that every year he would splash out on a "really over-the-top Christmas gift" to compensate. Lowballing female candidates is still rife, she said, noting that last year she rejected a job offer on pay grounds. A few months later, the same role was offered to a male peer at a significantly higher salary band. The switch to electronic trading might have blunted the banter-driven boys' club image, but foreign exchange is still male-dominated. One recruiter who specialises in currency trading said some companies pay extra commission if the shortlist for a job contains a woman, while others specify that a third of candidates need to be female. And yet few get hired. "Yes, it's difficult to find female talent for senior roles but, even when I show them a brilliant female candidate, they rarely ever get hired," said the recruiter. "I've given up, to be honest."



BOX 12.1 The corporate governance of banks (continued)

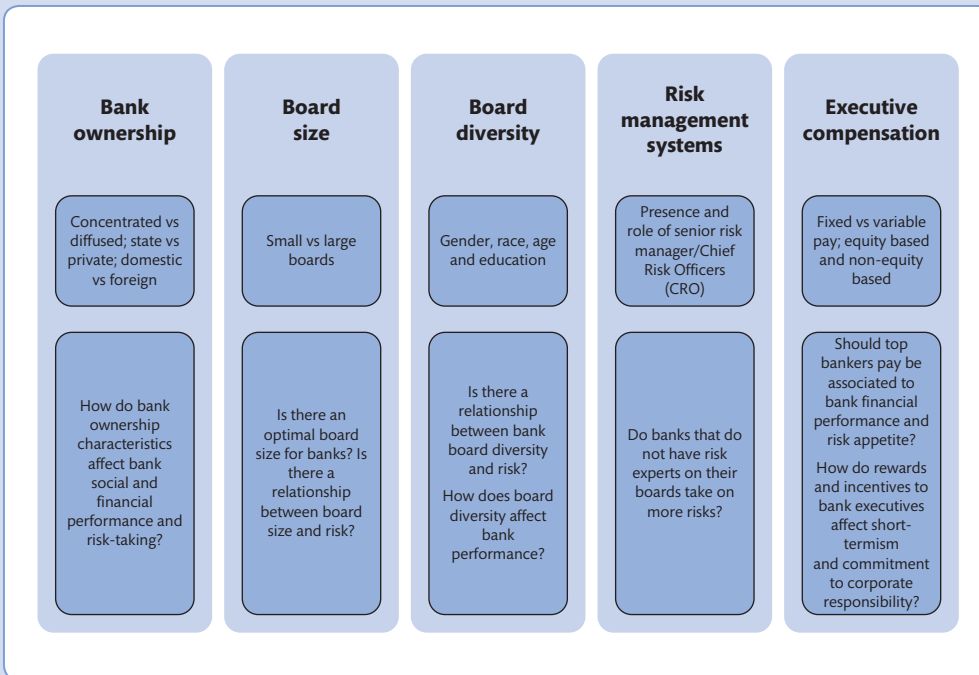


Figure 12.2 Corporate governance and risk in banking

In sum the corporate governance of banks remains a topical issue and over the last decade or so there has been an explosion of research looking at aspects relating to board features, the role of CEOs, executive pay and a whole host of related issues

that can, directly or indirectly, affect risk. Figure 12.2 summarises some of the key areas investigated for banks.

Key references: De Haan and Vlahu (2016); Hagendorff (2019); Jensen and Meckling (1976); Walker (2009).



Source: FT article from Szalay, E. (2019) 'Trading floor culture remains a barrier for senior female staff', *Financial Times*, 9 August.
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As mentioned in Chapter 11, the total risk a financial institution faces can be assigned to different sources. In the next sections we detail the management process of specific risks.

12.3 Credit risk management

Credit risk, defined as the potential that a bank borrower or counterparty will fail to meet its obligations in accordance with agreed terms, is the most familiar of banking risks (and it still remains the most difficult to quantify).

According to the BCBS (2000), the goal of credit risk management is 'to maximise a bank's risk-adjusted rate of return by maintaining credit risk exposure within acceptable parameters'. Banks need to manage the credit risk arising both from individual creditors and

individual transactions and the risk inherent in their entire portfolio. Furthermore, banks need to consider the relationships between credit risk and other risks. The effective management of credit risk is a critical component of a comprehensive approach to risk management and essential to the long-term success of any banking organisation.

While financial institutions can face difficulties for a number of reasons, in normal times loans that are not repaid (referred to as non-performing loans, or bad debts, or loan-losses) are the most frequent cause of bank losses.

For most banks, loans are the largest and most obvious source of credit risk; however, other sources of credit risk exist throughout the activities of a bank, both on and off the balance sheet. Traditionally, banks have monitored credit risk through a number of standard procedures, such as ceilings placed on the amount lent to any one customer and/or customers within a single industry and/or customers in a given country. While such procedures have long been a central feature of bank lending, credit risk measurement does raise several important issues:

- The size of the loan is not sufficient to measure the risk because risk has two dimensions – the quantity of risk, or the amount that can be lost, plus the quality of the risk, which is the likelihood of default. The quality of risk is often appraised through some form of credit ratings. These ratings may be internal to a bank or external when they come from a credit-rating agency. Measuring the quality of risk ultimately leads to quantifying the default probability of customers, plus the likelihood of any recovery (how much of the loan or other debt can be recovered) in the event of default. The probability of default is obviously not easy to quantify. Historical data on defaults by credit-rating class and/or by industry are available, but often they cannot be easily assigned to individual customers. The extent of recoveries is also unknown. Losses may depend upon guarantees, either from third parties or from any posted collateral, of recovery after bankruptcy and the liquidation of assets.
- The cumulated credit risk over a portfolio of transactions, either loans or market instruments, is difficult to quantify because of diversification effects. If the defaults of all customers tend to occur at the same time, the risk is much more significant than if those default events are not related (or independent). All banks, of course, protect themselves against risk through diversification, which makes simultaneous defaults very unlikely. However, the quantitative measurement of the impact of diversification still remains a modelling challenge.

Banks are increasingly facing credit risk (or counterparty risk) in various financial instruments other than loans, including acceptances, interbank transactions, trade financing, foreign exchange transactions, financial futures, swaps, bonds, equities, options, in the extension of commitments and guarantees, and the settlement of transactions.

Market transactions also generate credit risk. For instance, the inability of a company to service a swap, futures or options agreement, or make dividend repayments on bonds is also regarded as a credit risk. The loss in the event of default depends on the value of these instruments and their liquidity. If the default is totally unexpected, the loss is the market value of the instruments at the time of default. If the credit standing of the counterparty falls, for example Standard & Poor's reduces the credit rating of a counterparty from AAA to AA, it will still be possible to sell their instruments in the market at a discount. For financial instruments with more limited marketability such as over-the-counter (OTC) transactions, for example, as swaps and options, sale is not usually feasible. The credit risk of these types of instruments changes constantly with market movements during their lifespan. Therefore, the potential values of the transactions during the period of the contracts are at risk. Clearly,

there is a relationship between credit risk and market risk during this period because values depend on market movements.

Over the last decade banks have been seeking ways to measure credit risk more accurately, a need that has recently been strongly driven by a variety of factors:

- the growth of securitised loans and the secondary loan trading market (see also Chapter 18);
- the evolution of credit derivatives business;
- the increased emphasis on risk-adjusted performance measurement systems (where performance for different parts of the firm/bank is assessed relative to the risk taken and capital backing) and the desire to transfer credit risk;
- the desire of companies to manage the risk/return characteristics of their debt funding more effectively.

These factors have all contributed to the development of an increasingly more liquid and transparent market in tradable bank loans and other credit instruments. Such a market enables banks and large firms to trade their credit risk more effectively and therefore improve returns. It also provides banks with greater flexibility in meeting client needs as loan portfolios can be restructured in a more efficient manner, thereby releasing resources to be directed to higher-demand (and more profitable) areas. The growth in credit market instruments and the demand from banks to better assess the risks associated with their credit business has led to the development of various modelling techniques, similar in many respects to those previously developed to calculate market risk (such as JP Morgan's CreditMetricsTM or Credit Suisse First Boston Credit Risk+TM).²

Undoubtedly the 2007–2009 financial crisis has proved that these developments, particularly the growth in securitised lending, off-balance-sheet business and the increased reliance on external assessments by rating agencies in managing credit risk, resulted in banks' exposures being seriously understated. Typically, banks used complex quantitative modelling techniques to assess the risks associated with their credit business and indeed for their trading strategies too. One of the main problems, in this context, is that these models largely relied on 'normal' distributions which, although realistic in periods of prolonged stability and moderate growth (as in the *Great Moderation* era, between the late 1980s and 2007), can have devastating consequences in case of an unexpected (negative) shock to the system. It is possible that banks' over-reliance on sophisticated risk management processes increased moral hazard and encouraged banks to take on higher exposures as a result.

In a provocative speech, a former member of the executive Board of the ECB, Mr González-Páramo (2010), emphasised that one of the causes of the severity of the global financial crisis has been banks' deviation (or even abandonment in some cases) from some well-established rules in the management of risk, and particularly credit risk.³ These are common-sense risk management practices such as 'know your counterparties', 'invest only in products you understand', 'do not outsource credit risk management by relying exclusively on external credit assessments', 'do not rely exclusively on quantitative models, however sophisticated'. Again, as we discussed in Box 11.1, the development of an appropriate credit culture within banks is crucial in ensuring a more stable and reliable banking sector.

² See Bessis (2015), for an overview of the main features of credit risk models.

³ 'The challenges of credit risk management: Lessons learned from the crisis', speech available at www.bis.org/review/r100528d.pdf

The BCBS (2000) document 'Principles for the management of credit risk' still forms the backbone of present credit risk regulation. It identifies sound practices in the management of credit risk and specifically addresses the following areas:

- 1 establishing an appropriate credit risk environment;
- 2 operating under a sound credit-granting process;
- 3 maintaining an appropriate credit administration, measurement and monitoring process; and
- 4 ensuring adequate controls over credit risk.

Although specific credit risk management practices may differ among banks depending upon the nature and complexity of their credit activities, a comprehensive credit risk management programme will address these four areas. These practices should also be applied in conjunction with sound practices related to the assessment of asset quality, the adequacy of provisions and reserves, and the disclosure of credit risk.

12.4 Managing the lending function

Managing credit risk in retail banking, although based on the same broad principles, differs from wholesale banking in several ways. One first, obvious difference concerns the size of the loan commitments. Bad corporate loans can be very serious for banks because of the vast sums of money involved. Big corporate failures in the early 2000s have caused more than a serious headache to banks. High-profile failures such as Enron, WorldCom and a group of other big, mainly US telecoms and energy companies have left banks with a large amount of bad loans. Past crises in Latin America and Asia have also led to substantial write-offs. The global crisis of 2007–2009 has had major consequences on the balance sheets and performance of non-financial corporate businesses, and this affected significantly the cash flow and capital of the banking sector. Retail lending, however, while unlikely to create serious consequences for a bank in case of individual loan defaults (although it can still create problems if a large number of retail loans default, say, as a result of a bank being overexposed to a particular sector of the economy, particularly real estate), is more difficult to assess because of the lack of information on the creditworthiness of individual potential borrowers.

12.4.1 Retail lending

An accurate credit decision, given a bank's credit standards, is the one that maximises the value of the loan for the bank and minimises the risk of default. Consequently, gathering, processing and analysing information on potential borrowers are key steps in credit risk management.

Prior to making a lending decision, banks need to assess the risk–return trade-off of a loan; this process involves both an assessment of the risk of the applicant and the applicant's business, an analysis of the external environment, the purpose of the loan and the particular loan

structure requested by the applicant. One key step in this process is pricing the loan, where the 'price' (**loan rate**), assuming there are no other costs, should be:

$$R^L = \frac{1 + r}{1 - d} - 1 \quad (12.1)$$

where:

R^L = profitable loan rate

r = risk-free interest rate (i.e. the rate of return on a 'risk-free' investment, such as government bonds)

d = expected probability of default.

Equation (12.1) shows that the profitable loan rate increases with the borrower's expected probability of default. From a lenders' perspective, the gross return on the loan should take into account several key factors:

- interest rate on the loan;
- fees relating to the loan;
- credit risk premium on the loan;
- collateral backing of the loan; and
- other non-price terms (e.g. clauses and conditions on the use of the loan).

Following a general model along the lines developed by Saunders and Cornett (2017), the gross return on a loan is:

$$1 + k = 1 + \frac{f + (L + M)}{1 - [b(1 - R)]} \quad (12.2)$$

where:

k = contractually promised gross return on the loan per £ lent

f = administration fee

L = base lending rate

M = credit risk premium on the loan

b = compensating balance requirement (as a percentage of the loan)

R = reserve requirement.

L in formula (12.2) reflects the bank's marginal cost of funds (or the so-called prime lending rate in the United States – the interest rate a bank charges its best or 'prime' customers). Compensating balances (b) is the portion of a loan that a borrower may be required to hold as deposit at the bank. They are commonly used in corporate lending business. Reserve ratios (R) are effectively a tax on deposits, and can cause the loan price to rise as banks are not earning income for every £ of reserves. Finally, the loan price should include a credit risk premium (M) and an administration fee (f). The former should reflect the risk profile of the borrower: the riskier the borrower, the higher the premium. The latter should cover all the costs incurred in the origination and administration of the loan.

Another factor that influences the price of the loan is the presence of any collateral (assets backing the loan that can include such things as residential property, other real estate, securities, etc.): the rate charged should be lower than in the case of no collateral. However, in times of economic trouble the price of collateral can become very volatile and banks may have to raise either the loan price or the amount of collateral required. If collateral values fall dramatically there can be a banking sector collapse – as was the case in the Japanese banking system crises that occurred in 1997–1998 (discussed in Chapter 16). Obviously, if the value of the collateral is linked to the ability of the borrower to repay; a decrease in the value of the collateral will increase the probability of default. Box 12.2 illustrates how the fall in UK property prices in the early 1990s and in the aftermath of the 2007–2009 crises resulted in loan losses and negative equity.

BOX 12.2 MORTGAGE MARKET, EQUITY WITHDRAWAL AND NEGATIVE EQUITY

Over the past 40 years or so the UK housing market has experienced persistent volatility (rapid fluctuations in house prices) that resulted in major economic and social costs. During the late 1980s, UK house prices soared, boosting people's confidence in the housing market and prompting them to borrow against the value of their homes. Between 1980 and 1990, average mortgage debt in the UK more than doubled relative to income. A **mortgage equity withdrawal** relates to homeowners borrowing against the increased value of their property (capital gains) by taking out additional housing equity loans. The Bank of England's estimate of mortgage equity withdrawal (MEW) measures the part of consumer borrowing from mortgage lenders that is not invested in the housing market. Levels of mortgage equity withdrawal reached very high levels in the late 1980s; mortgage equity loans were a major factor behind the consumer boom of 1987–1988. However, the collapse in the housing market in the early 1990s led to over 1.5 million homeowners being in **negative equity** – a situation where the market value of their property was less than the outstanding mortgage loan. The slump in the UK housing market coincided with record increases in mortgage arrears and repossessions. Between 1989 and 1993, as both interest rates and unemployment rose, the number of households in mortgage arrears increased. Arrears peaked in 1992 when over 350,000 households owed six or more months' payments. Repossessions were highest in 1991 at 75,450, falling to 58,540 by 1993.

As shown in Figure 12.3, by the mid-2000s, following a period of sharp increases in housing prices

repossessions were relatively low by historical standards. However, the financial turmoil that started in 2007, and the global credit crunch and recession that followed, profoundly affected the UK housing market and repossessions peaked in 2009 at nearly 49,000 homes.

The volatility of the early 1990s and late 2000s caused distorted housing choices and increased risks. From the lenders' point of view these housing market problems resulted in costly losses to banks and building societies. Falling nominal house prices reduced the amount of equity in housing and provided incentives for borrowers to accumulate arrears and for lenders to repossess property that had been used to secure loans.

In March 2020 the UK government adopted a package of support measures for individuals and businesses affected by the Covid-19 pandemic. These included job-protection schemes, provisional changes to the social security system to protect, at least partially, household incomes, and mortgage-support measures. These latter measures were supposed to assist homeowners to manage their repayments and avoid mortgage arrears and repossessions. Mortgage lenders essentially agreed to support customers experiencing difficulties through a payment 'holiday' or deferral of up to three months from their monthly payments. In May 2020, this holiday period extended for a further three months; however, with the second national lockdown in autumn 2020, new measures to protect homeowners from repossessions were announced, including a moratorium on the enforcement of lender repossessions (Wilson *et al.*, 2020).



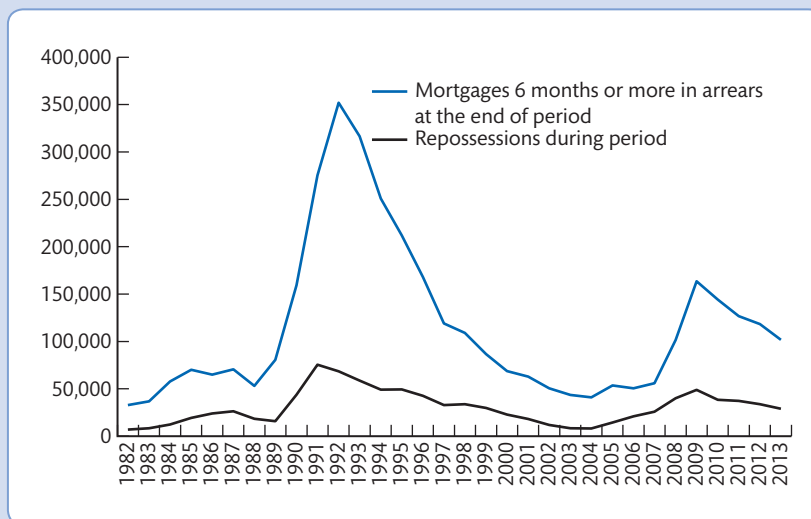
Box 12.2 Mortgage market, equity withdrawal and negative equity (*continued*)

Figure 12.3 Home repossessions in the UK

Source: UK Ministry of Housing, Communities & Local Government. Data from live table 1300 (discontinued in 2014).

Loans availability in retail markets may not be linked simply to the loan price, but restricted to a selected category of borrowers. This is a method for managing credit risk (**credit rationing**) that attempts to minimise the problem of adverse selection in loan markets. To reduce risk exposure, banks can limit the amount of credit available to a certain class of borrowers; for example, think of the credit limit on your credit card. Although it is illegal to discriminate against borrowers for reasons such as race, gender, religion, sexual orientation and address, there is no automatic 'right' to credit and people can be refused credit for a number of different reasons.

As we have seen, a loan evaluation process focuses on evaluating the prospective risk and return on a loan. There are several techniques or models for assessing credit risk, but they can be broadly divided in qualitative models and quantitative models. Qualitative models are normally used when there is limited available information on a borrower. Bank managers have to gather information from private sources and the amount of information needed will be proportional to the size of the loan. Quantitative models are to assess borrowers' credit-worthiness based on the estimated probability of default.

12.4.2 Credit checking and credit scoring

Lenders want to minimise the information asymmetry problems in the retail loan market and therefore aim to ensure that potential borrowers are a good risk and do not have a history of bad debts and unpaid loans. To do this they will do two things: **credit checking** and **credit scoring**.

12.4.2.1 Credit checking

Lenders will check the applicant's entry on credit registers. **Credit reference agencies** in the UK such as Experian, Equifax and CallCredit hold factual information on retail customers and this allows a lender to check individuals' names and address and past credit history, including any county court judgments or defaults recorded against the individual.⁴ This process will provide a person's so-called *credit reference*.

12.4.2.2 Credit scoring

To obtain information on a potential borrower, banks will initially adopt a qualitative approach, which involves asking the applicant a number of questions. They will then allocate points (weights) to the answers. Questions may concern the applicant's employment history, the length of time as a customer of the bank, the number and type of accounts held, the length of time at their present address and so on. Personal judgement on behalf of loan officers based on the 'five Cs' (character, capacity, capital, collateral and conditions) is now commonly replaced by a quantitative approach based on the use of the information provided by the applicant to calculate the probability of default.⁵ Using statistical programmes, creditors compare the information to the credit performance of consumers with similar profiles. A credit-scoring system awards points for each factor that helps predict who is most likely to repay a debt. A total number of points – a credit score – helps predict how creditworthy the applicant is, that is, how likely it is that they will repay a loan and make the payments when due. Lenders will never divulge how their credit scoring works for fear of fraud and each lender will have their own system. The fact that your loan application has been turned down by one lender does not necessarily imply that it will also be declined by other lenders.

Although such a system may seem arbitrary or impersonal, when it is properly designed it can help make decisions faster, more accurately and more impartially than individuals. In marginal cases, applicants are referred to a credit manager who decides whether the company or lender will extend credit. This may allow for discussion and negotiation between the credit manager and the consumer. Box 12.3 illustrates how a credit-scoring system works in practice.

BOX 12.3 CREDIT SCORING: HOW TO SCORE HIGH

- Have you paid your bills on time?

Payment history typically is a significant factor. It is likely that your score will be affected negatively if you have paid bills late, had an account referred to collections, or declared bankrupt, if that history is reflected on your credit report.

⁴ Experian and Equifax are also among the major national *credit bureaus* in the US, together with TransUnion.

⁵ The so-called five Cs of a credit decision can be described as follows: (1) *character*, which refers to the willingness of the borrower to repay the loan; (2) *capacity*, which refers to the borrower's cash flow and the ability of that cash flow to service that debt; (3) *capital*, which refers to the strength of the borrower's balance sheet; (4) *collateral*, which refers to the security backing up the loan; and finally, (5) *conditions*, which refer to the borrower's sensitivity to external forces such as interest rate and business cycles and competitive pressures (Sinkey, 2002; Saunders and Cornett, 2017).

Box 12.3 Credit scoring: how to score high (continued)

- What is your outstanding debt?

Many scoring models evaluate the amount of debt you have compared to your credit limit. If the amount you owe is close to your credit limit, it is likely that it will have a negative effect on your score.

- How long is your credit history?

Generally, models consider the length of your credit track record. An insufficient credit history may have an effect on your score, but that can be offset by other factors, such as timely payments and low balances.

- Have you applied for new credit recently?

Many scoring models consider whether you have applied for credit recently by looking at 'inquiries' on your credit report when you apply for credit. If you have applied for too many new accounts recently, it is likely that it may negatively affect your score. However, not all inquiries are counted. Inquiries by creditors who are monitoring your account or looking at credit reports to make 'pre-screened' credit offers are not counted.

- How many and what types of credit accounts do you have?

Although it is generally good to have established credit accounts, too many credit card accounts may have a negative effect on your score. In addition, many models consider the type of credit accounts you have. For example, under some scoring models, loans from finance companies may negatively affect your credit score.

Source: Adapted from the Federal Trade Commission (www.consumer.ftc.gov).

Credit scoring can be applied both to individuals and to corporations; obviously the variables used to define the scoring system will differ. Sinkey (2002) and Saunders and Cornett (2017) provide detailed overviews of the main traditional credit-scoring models. In recent years, new lending models based on predictive machine learning techniques have emerged, such as credit scoring based on **digital footprint** data. These models use individuals' online behaviour to predict default and although they are gaining popularity, they give rise to other risks that have not yet been completely evaluated by society. For example, the algorithm used in these models could be biased and perpetuate inequalities if the data contains stereotypical concepts of gender and race.

12.4.2.3 Linear probability models

Loans are divided in two groups, those that defaulted ($Z_i = 1$) and those that did not ($Z_i = 0$). These observations are then regressed on a set of n variables reflecting quantitative information about the i th borrower:

$$Z_i = \sum_{j=1}^n \beta_j X_{ij} + \epsilon \quad (12.3)$$

Logit (and Probit) models

These constrain the cumulative probability of default on a loan between zero and one and assume the probability of default to be logistically distributed (or have a normal distribution in the Probit case).

Linear discriminant models

These models (which include the Altman Z-score model) divide borrowers according to their derived Z-scores into high or low default risk classes, contingent on their observed characteristics (X_j):

$$Z_i = \sum_{j=1}^n \alpha_j X_{ij} \quad (12.4)$$

12.4.3 Managing the loan portfolio

Moving from individual loans to a bank's loan portfolio, the first step in credit risk management is diversification. The principle behind portfolio diversification is the same as the one behind the old saying 'don't put all your eggs in one basket'. Bank managers should diversify their lending to different sectors of the economy, different geographical locations, different types of industry, different maturities and so on. By diversifying their loan portfolios, i.e. by owning assets whose returns are not statistically correlated, banks can reduce the impact of any failure by diversifying away the unsystematic risk. A heavy concentration of loans in one sector of the economy can cause serious trouble for banks. For example, in the UK, building societies have historically concentrated the majority of their lending to finance residential property purchase and have therefore suffered a high number of non-performing loans when the housing market collapsed (see Chapter 8 and Box 12.2). If loans are not correlated, banks can expect to increase the expected returns on their loan portfolio by diversifying across asset classes. In other words, it is possible to apply standard portfolio theory to obtain a measure of aggregate credit risk exposure.⁶

When assessing the credit risk of the aggregate loan portfolio, bank managers need to calculate the following:

- the expected loss, for each loan and for the whole portfolio, over a specific time horizon;
- the unexpected loss for each loan and for the whole portfolio (i.e. the volatility of loss);
- the probability distribution of credit loss for the portfolio and the capital requirement for a given confidence level and time horizon (see Figure 12.4).

There are three factors that drive expected and unexpected losses on a credit portfolio:

- customer default risk – determined by the risk-grade profile of the portfolio;
- exposure – the amount that is likely to be outstanding at the time of default;
- loss given default – determined by the level of security cover, the effectiveness of the recovery process and the credit cycle.

The calculation of the expected loss is based on the current risk profile of the portfolio, possibly ignoring historical loss rates. Banks that rely on their average loss experience to derive expected loss are assuming that the risk profile, business mix and risk management processes remain constant over time. However, the existing profile might be considerably better than the one that created the losses in the past.

⁶For an introduction to portfolio theory, see Appendix A2.

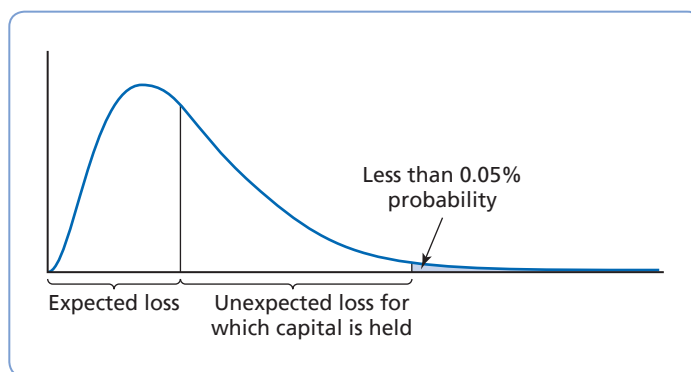


Figure 12.4 Probability distribution of portfolio losses

The main problem with applying portfolio theory to banks' loan portfolios is that, in the vast majority of cases, bank loans are non-tradable assets. As we noted earlier, there are several products in the market that attempt to deal with these issues. The best known are Credit Suisse First Boston (CSFB) Credit Risk+™, CreditPortfolioView™ by McKinsey and JPMorgan's CreditMetrics™. In addition, firms such as KMV⁷ and KPMG are actively participating in the debate and are openly sharing many of their analytical engines. KMV was taken over by Moody's in 2002 and since then this portfolio model is called M-KMV. Moody's has several products on the market, including Credit Monitor™ and Portfolio Manager™. KPMG's contribution is the Loan Analysis System™.⁸

12.5 Managing interest rate risk

Interest rate risk can be defined as the exposure of a bank's financial condition to adverse movements in interest rates (BCBS, 2004). For banks, interest rate risk derives from mismatching the maturities of assets and liabilities, as part of their asset transformation function. Banks use funds raised in the form of short-term deposits to issue or buy longer-term assets (loans and bonds) but typically their prices and income react differently to interest rate fluctuations compared to short-term prices and interest expenses.

Interest rate risk management is concerned with the management of the interest rate exposure of a bank. Traditionally, it is managed within the ALM function but, given the increased volatility of interest rates, together with the increased interest rate risk arising from off-balance-sheet activities, it is now often managed by a dedicated department. As seen in Section 11.3, interest rate risk arises from potential variations in banks' returns that derive from unanticipated changes in interest rates. Changes in interest rates also affect a bank's underlying economic value. The value of a bank's assets, liabilities and off-balance-sheet business is affected by a change in rates because the present value of future cash flows, and in some cases the cash flows themselves, are altered.⁹ A bank interest rate risk exposure refers

⁷ KMV is the acronym for Kealhofer, McQuown and Vasicek, the originators of the KMV methodology.

⁸ For more information on these products, see Credit Suisse First Boston (1997) and Bessis (2015).

⁹ This occurs, for example, when the bank funds a one-year loan then reprices monthly based on the one-month T-bill rate, with a one-year deposit that reprices monthly based on one-month LIBOR. Interest rate changes can result in unexpected changes in the cash flows and earnings spreads between assets and liabilities. This type of interest rate risk is referred to as 'basis risk'.

to market value changes in its equity position as a result of unexpected changes in market interest rates. When attempting to measure and manage interest rate risk, it is important to note that the exposure to such risk concerns *future* losses (or gains) and therefore some uncertainty will always be present. Some interest rate management techniques involve a forecast of possible interest rate scenarios and, as in every attempt to forecast the future, there is no such thing as 100 per cent accuracy. In other words, managing interest rate risk is about identifying, measuring and controlling such risk, taking uncertainty into account.

There are two broad management approaches that are used to measure interest rate risk and these are known as 'gap' and 'duration' analysis.

12.5.1 Gap analysis

Gap analysis is possibly the best-known interest rate risk management technique. The 'gap' (Figure 12.5) refers to the difference between interest rate-sensitive assets and interest rate-sensitive liabilities over a specific time horizon. If the interest rate-sensitive liabilities are greater than the interest rate-sensitive assets, then an increase in interest rates will reduce a bank's profit and vice versa. In the basic gap analysis, the focus is on the maturity of the rate-sensitive assets and liabilities:

$$\text{GAP} = \text{RSA} - \text{RSL} \quad (12.5)$$

where:

RSA = rate-sensitive assets

RSL = rate-sensitive liabilities.

An asset or a liability is defined as rate sensitive if the cash flow from the asset or liability changes in the same direction as changes in interest rates. The gap ratio in equation (12.5) is also called the *interest-sensitivity ratio*. If this ratio is equal to one, then the rate

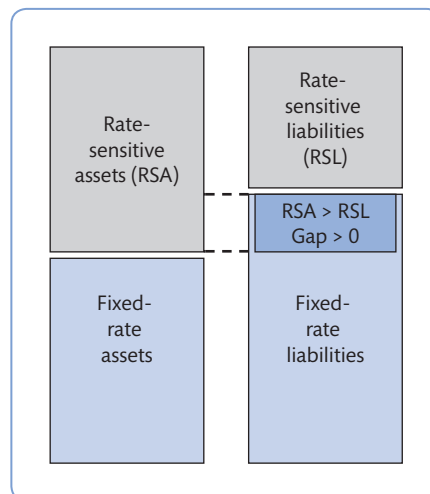


Figure 12.5 Rate sensitivity and positive gap

sensitivity of assets and liabilities is perfectly matched. However, most banks have a positive gap ($RSA > RSL$) since they borrow short and lend long, and therefore have assets that will mature later than liabilities. The main aim of gap analysis is to evaluate the impact of a change in interest rates on the bank net interest income (NII) and net interest margin (NIM). Ideally, the gap should be managed in such a way as to expand when interest rates are rising and contract when interest rates are declining. However, it is difficult for bank managers to know what phase of the interest rate cycle they are facing. Furthermore, bank customers may be seeking opposite interest rate positions compared to the bank.

Up to now, we have defined the gap as being related to (or a function of) a specified time horizon (for example, 90 days). However, this is rather arbitrary, as it does not clearly indicate what time period is appropriate for determining the interest rate sensitivity of assets and liabilities. For instance, focusing on a short-term gap may ignore reinvestment risk (the risk that loans are repaid early). One extension of the basic gap model is the *maturity bucket* approach. Each of the bank assets and liabilities is classified according to its maturity and placed into 'maturity buckets', for example overnight, 3 months, 3–6 months and so on. Analysts compute both incremental and cumulative gap results. An incremental gap is defined as $RSA - RSL$ in each time bucket; the cumulative gap is the cumulative subtotal of the incremental gaps.

As illustrated in Table 12.2, as total assets equal total liabilities by definition, the incremental gaps must total to zero and therefore the last cumulative gap must be zero. The maturity bucket approach allows bank managers to concentrate on the cumulative gaps for the different time buckets.

One extension is the maturity gap (M Gap):

$$M\text{ Gap} = W_A RSA - W_L RSL \quad (12.6)$$

where:

$W_A RSA$ = weighted average rate-sensitive assets

$W_L RSL$ = weighted average rate-sensitive liabilities.

This model better reflects the economic reality or the true value of assets and liabilities if the bank portfolio was liquidated at today's prices. If the maturity of a bank's assets is greater than the maturity of its liabilities, an increase in interest rates will cause the value of the assets to fall more than the value of the liabilities because the assets mature later. The bigger the maturity gap, the more a bank's net worth will suffer by an increase in interest rates.

Table 12.2 Maturity bucket gap (£mil)

	Assets	Liabilities	Gap	Cumulative gap
One day	40	30	+10	-10
More than 1 day less than 3 months	50	60	-10	0
More than 3 months less than 6 months	90	110	-20	-20
More than 6 months less than 12 months	110	120	-10	-30
More than one year less than 5 years	80	70	+10	-20
Over 5 years	30	10	+20	0
	400	400		

	Gap	Maturity
If interest rates ↑	RSA > RSL	Assets < Liabilities
If interest rates ↓	RSA < RSL	Assets > Liabilities

Figure 12.6 Strategic matrix for profit maximisation

Figure 12.6 reports a simple summary for gap and maturity models in the form of a strategic matrix to maximise banks' earnings (in terms of NII and NIM) when interest rates fluctuate.

Another extension is the so-called '*dynamic gap analysis*' approach, which involves forecasting interest rate changes and expected changes in the bank's balance sheet for several periods in the future. Software models provide bank managers with simulation tools to inform them of the way in which gaps are expected to be structured at certain times in the future.

Gap analysis was one of the first methods developed to measure a bank's interest rate risk exposure, and continues to be widely used by banks. Despite the extensions, the gap model has been defined as 'naïve' and has been subject to a number of criticisms as the approach:

- fails to take into account the market value effect (i.e. the new value of the asset given changes in interest rates);
- suffers from over aggregation, that is, it fails to consider for intra-bucket effects;
- fails to deal with run-offs, which is the periodic cash flow of interest and principal amortisation payments on long-term assets;
- ignores banks' exposure to pre-payment risk (the risk that loans will be repaid early);
- ignores differences in spreads between interest rates that could arise as the level of market interest rates changes (basis risk);
- does not take into account any changes in the timing of payments that might occur as a result of changes in the interest rate environment;
- generally oversimplifies the complexity of a bank's ALM.

For these reasons, gap analysis provides only a rough approximation of the actual impact of changes in interest rates.

12.5.2 Duration analysis

Duration is a measure of the average life of an asset's (or liability's) cash flow. **Duration analysis** takes into account the average life of an asset (or liability) rather than its maturity. It is a technique borrowed from bond portfolio management, where duration

is defined as a weighted average of the maturities of the individual coupon payments. In this context, duration may be different from maturity if, for example, an asset repayment schedule includes interest and principal. A three-year car loan that is repaid with monthly instalments will have duration different from its maturity. Maturity and duration are only ever equal in the case of single payment assets and zero coupon bonds. Higher duration implies that a given change in the level of interest rates will have a larger impact on economic value. The duration of a coupon bond is expressed by the formula (known as Macaulay duration):

$$D = 1 * \frac{C_1/(1+Y)^1}{V} + 2 * \frac{C_2/(1+Y)^2}{V} + \dots + n * \frac{(C_n + P_n)/(1+Y)^n}{V} \quad (12.7)$$

where:

Y = the bond's internal yield or yield-to-maturity (YTM)

C_1 = annual coupon payment in year 1

P_n = principal payment

n = number of years to maturity

V = current market value of the bond.

The formula provides the weighted average payment stream, where the maturity of each payment is weighted by the fraction of the total value of the bond accounted for by the payment. As can be seen from the example in Box 12.4, the emphasis in duration analysis is on the market value rather than on the book value, as was the case in gap analysis.

Using formula (12.7) it is possible to compute the duration of the entire asset and liability portfolios of a bank. By matching the duration of assets and liabilities, movements in interest rates should have roughly the same effect on both sides of the balance sheet.

BOX 12.4 EXAMPLE OF MACAULAY DURATION

Consider a bond with the following characteristics:

- £100 annual coupon;
- 2 years to maturity;
- YTM = 10 per cent;
- market value £1,000.

The Macaulay duration for this bond is 1.909 years:

$$1.909 = 1 * \frac{£100/(1.1)}{£1000} + 2 * \frac{£1100/(1.1)^2}{£1000}$$

Duration gap (DG) measures the mismatch between the duration of a bank's assets and its liabilities:

$$DG = \left(D_A - \frac{L}{A} D_L \right) \quad (12.8)$$

where:

A = market value of assets

L = market value of liabilities

D_A = duration of assets

D_L = duration of liabilities

L/A = leverage or gearing ratio.

The impact of a change in interest rates on the value of a bank's equity can be calculated as follows:

$$\Delta E = -DG \left(\frac{\Delta r}{(1 + r)} \right) A \quad (12.9)$$

where:

ΔE = change in the value of bank equity

DG = duration gap

Δr = change in interest rate

A = market value of assets.

Estimates derived from a standard duration approach may provide an acceptable approximation of a bank's exposure to changes in economic value for relatively non-complex banks. However, there are a number of problems arising from the use of the duration measure:

BOX 12.5 EXAMPLE OF DURATION GAP

Consider a bank with the following characteristics:

- £500 million of assets;
- £400 million of liabilities;
- £100 million of own equity;
- the duration of assets is five years;
- the duration of liabilities is three years.

Let's suppose that bank management expect an interest rate increase of 0.25 per cent to 4.5 per cent following the next meeting of the Bank of England's Monetary Policy Committee (MPC).

$$DG = [5 - (400/500)3] = 2.6$$

$$\Delta E = -2.6 [0.0025/(1 + 0.0425)] 500 = -3.12 \text{ million}$$

In this case, an increase in interest rates from 4.25 to 4.5 per cent will decrease the equity value by £3.12 million.

- Convexity – the duration formula implies a linear relationship between changes in interest rate and changes in bond price; in reality the relationship is convex. Duration is a good approximation for small changes but it becomes less accurate for larger changes.
- Data requirements – the calculation of the duration gap can be data demanding.
- Duration generally focuses on just one form of interest rate risk exposure – re-pricing risk – and ignores interest rate risk arising from changes in the relationship among interest rates within a time-band (basis risk).
- The simplifying assumptions that underlie the calculation of standard duration mean that the risk from off-balance-sheet activities may be underestimated.

12.5.3 Simulation approaches

Many large banks employ more sophisticated interest rate risk measurement systems than those based on simple maturity/re-pricing schedules. These simulation techniques typically involve detailed assessments of the potential effects of changes in interest rates on earnings and economic value by simulating the future path of interest rates and their impact on cash flows. Simulation approaches typically involve a more detailed breakdown of various categories of on- and off-balance-sheet positions, so that specific assumptions about the interest and principal payments and non-interest income and expense arising from each type of position can be incorporated. In addition, simulation techniques can incorporate more varied and refined changes in the interest rate environment, ranging from changes in the slope and shape of the yield curve to interest rate scenarios derived from (relatively complex) statistical approaches such as Monte Carlo simulations.¹⁰

We can distinguish between:

- *static simulations*, where only the cash flows arising from the bank's current on- and off-balance-sheet positions are assessed; and
- *dynamic simulations*, where the model builds in more detailed assumptions about the future course of interest rates and the expected changes in a bank's business activity over that time.

The usefulness of simulation-based interest rate risk measurement techniques depends on the validity of the underlying assumptions and the accuracy of the basic methodology. In its document 'Principles for the management of interest rate risk', the BCBS warns that the output of sophisticated simulations must be assessed in the light of the validity of the simulation's assumptions about future interest rates and the behaviour of the bank and its customers (BCBS, 1997b). One of the primary concerns of BCBS is that such simulations could become 'black boxes' that lead to false confidence in the precision of the estimates.

Subsequent recommendations of the Federal Reserve Board (2010) included the need to run dynamic earning simulations together with static ones in order to have a complete profile of the bank's interest rate risk, as well as the use of economic value-based methodologies to broaden the assessment in quantifying interest rate risk exposures and overcome the limitations of static and dynamic models. Specifically, the economic-value approaches focus on a longer time horizon and capture all future cash flows expected from current assets and liabilities. They are also regarded as more effective in considering embedded options in a typical institution's portfolio.

¹⁰ A Monte Carlo simulation is a computerised technique which is the basis for probabilistic risk analysis, and which replicates real-life occurrences by mathematically modelling a projected event.

To summarise:

- *Economic value of equity (EVE) models* measure the degree to which the economic values of a bank's positions change under different interest rate scenarios and can include dynamic technique to provide forward-looking estimates of economic value.

Overall, the most recent guidelines encourage banks to use a range of measurement approaches. The aim is to ensure that the interest rate risk measurement system captures all on- and off-balance-sheet positions and incorporates stress-testing processes to identify and quantify the institution's exposure and potential problem areas.

12.6 Managing liquidity risk

As defined in Section 11.4, a bank faces liquidity risk when, because of lack of confidence or unexpected need for cash, withdrawals are higher than normal and the bank is unable to meet its liabilities. Sound liquidity management can reduce the probability of serious problems. The importance of liquidity goes beyond the individual bank, since a liquidity shortfall at a single institution can have system-wide repercussions. Systemic risk and the problems of contagion and bank runs have been discussed already in Chapter 7 and elsewhere in this text.

If a bank experiences a temporary liquidity problem, and it is either unable or unwilling to borrow on the interbank market, the central bank can provide funds, in the form of loans and advances. However, central bank borrowing is costly not only in terms of the interest rates charged but also in terms of the bank's reputation.

Liquidity pressures can arise from both sides of the balance sheet. On the liability side, unexpectedly high cash withdrawals can cause solvent banks to have liquidity problems. On the asset side, liquidity problems can be caused by unexpectedly high loan defaults and by customers unexpectedly drawing down lines of credit. Liquidity pressures can arise from off-balance-sheet activities as well as from problems in the payment system. Contingent liabilities, such as letters of credit and financial guarantees, represent a potentially significant drain of funds for a bank, but are usually not dependent on a bank's liquidity position. Other potential sources of cash outflows include payments relating to transactions involving swaps, OTC options, other interest rate and forward foreign exchange rate contracts, margin calls and early termination agreements.

Liquidity management is an integral part of the ALM function. Liquidity risk management aims at protecting a bank against liquidity risk, that is, to avoid a situation of negative net liquid assets.

To avoid liquidity problems, a bank can hold liquid assets. However, increased liquidity comes at a cost. There is a trade-off between liquidity and profitability, as the more liquid the asset, the lower the rate of return. Instead of holding liquid funds, a bank could make more profitable loans. Despite the costs, however, the holding of liquid assets is necessary as it:

- reassures creditors that the bank is safe and able to meet its liabilities;
- signals to the market that the bank is prudent and well managed;
- ensures that all lending commitments can be met;
- avoids forced sale of the bank's assets;

- avoids having to pay excessive borrowing costs in the interbank markets; and
- avoids central bank borrowing.

Banks can minimise withdrawal risk by diversifying funding sources (liability management). Prudent banks will also seek to minimise their *volatility ratio*:

$$VR = (VL - LA)/(TA - LA) \quad (12.10)$$

where:

VL = volatile liabilities

LA = liquid assets

TA = total assets

Prudent banks will have a volatility ratio lower than zero.

In measuring and managing a bank's liquidity exposure, the following techniques may be used:

- cash flow projections of daily liquidity positions;
- cash flow projections of daily liquidity sources;
- scenario analysis and simulation models;
- liquidity gap analysis.

12.6.1 Liquidity gap analysis and financing gap

Liquidity gap analysis is the most widely used technique for managing a bank's liquidity position. As we have already discussed, liquidity risk is generated in the balance sheet by a mismatch between the size and maturity of assets and liabilities. It is the risk that the bank is holding insufficient liquid assets on its balance sheet to meet requirements. The liquidity gap is defined as the difference between net liquid assets and unpredictable (or volatile) liabilities. If a bank's net liquid assets are less than liabilities then the bank needs to purchase funds in the market to fill the shortfall in liquid assets. Banks typically will examine the maturity profile of their assets and liabilities to identify mismatches in liquidity that require funding.

$$L \text{ Gap} = NLA - VL \quad (12.11)$$

where:

NLA = net liquid assets

VL = volatile liabilities

The liquidity gap analysis is similar to the *maturity bucket approach* we have discussed for interest rate risk management. In this case, balance sheet items are placed in a bucket according to the expected timing of cash flows. Net mismatched positions are accumulated over time to produce a cumulative net mismatch position. In this way, the bank can monitor the amount of cash that becomes available over time.

Another useful measure of bank liquidity is the financing gap (F Gap):

$$\text{F Gap} = \text{Average loans} - \text{Average deposits} \quad (12.12)$$

If the F Gap is positive, the bank needs cash and will have either to sell some assets or borrow on the interbank market. The bigger the F Gap, the more a bank needs to borrow and the greater its exposure to liquidity risk. However, technological and financial innovations have provided banks with new ways of funding their activities and managing their liquidity. A declining ability to rely on core deposits, together with increased reliance on wholesale funds, has changed the way banks view liquidity.

During the early stages of the sub-prime crisis in 2007 many banks, despite being adequately capitalised, suffered significant difficulties mainly due to the lack of prudent liquidity management. The crisis has shown that markets can become illiquid very quickly and that the illiquidity can last long periods. The Basel Committee responded by setting out key principles, detailed guidelines and a framework for effective liquidity risk management and supervision in banking and the related key documents have now been integrated into the consolidated Basel Framework. The two fundamental minimum standards of the most recent Basel liquidity framework are the liquidity coverage ratio and the net stable funding ratio, explained in detail in Chapter 7 (Section 7.11.2).

12.7 Managing market risk

Market risk is the risk resulting from adverse movements in the level or volatility of market prices of interest rate instruments, equities, commodities and currencies (see Section 11.6). Market risk is usually measured as the potential gain/loss in a position/portfolio that is associated with a price movement of a given probability over a specified time horizon.

Financial institutions have always faced market risk; however, the sharp increase in asset trading since the 1980s has increased the need to ensure that these institutions have the appropriate management systems to control (and the capital to absorb) the risks posed by market-related exposures. As a risk, market risk gained a high profile when the Basel Committee on Banking Supervision published 'The supervisory treatment of market risks' (1993), in which for the first time it was proposed that market risk, in addition to credit risk, needed to be taken into account for the calculation of bank capital requirements (this updated the 1988 Basel Capital Adequacy Accord).

The 1993 BCBS consultative document put forward a standardised measurement framework to calculate market risk for interest rates, equities and currencies which differentiates capital requirements for specific risk from those for general market risk. The BCBS (1996) document 'Amendment to the capital accord to incorporate market risks' provides the first framework for capital charges relative to market risk. It sets forth two approaches for calculating the capital charge to cover market risks: *the standardised approach* and *the internal models approach*. The methodology for banks using the standardised approach is based on a 'building blocks' approach, in which the specific risk and the general market risk arising from securities positions are measured separately. Meanwhile, the focus of many internal models is on the bank's general market risk exposure, leaving specific risk (i.e. exposures to specific issuers) to be measured largely through separate credit risk measurement systems.

The central component of market risk management is **value-at-risk** (VaR), which is discussed in Section 12.7.1. Although VaR was originally developed to manage market risk, it has been extended to incorporate other risks, including credit risk. Post 2007–2009 crisis, the market risk framework was revised by the Committee in 2009 and adopted a revised framework (BCBS, 2010c), which became known as Basel 2.5¹¹ and included the following main reforms in the calculation of market risk capital:

- the *stressed value-at risk* (SVaR), which takes better account of tail risk (losses that banks can suffer in stressed market conditions);
- the *incremental risk charge* (IRC) framework, which determines capital requirements both for default risk and ratings **migration risk** (for unsecuritised credit products with issuer risk like bonds, CDs, equities, etc.);
- the exclusion of most trading book *securitisation exposures* from the internal models approach.

More recently, the BCBS also carried out a fundamental review of the trading book with the aim to improve the design and coherence of the capital standard for market risk. This resulted in a revised framework for market risk introduced in January 2016 (BCBS, 2016a) that: (i) set detailed criteria for the assignment of instruments to the trading book; (ii) improved the internal models approach to better deal with risks that were detected during the crisis; (iii) strengthened the supervisory approval process related to the use of internal models; and (iv) advanced a novel and more risk-sensitive standardised approach. Table 12.3

Table 12.3 Selected elements of internal and standard models approaches in the January 2016 revised market risk framework

Internal models approach	Standard model approach
• Single expected shortfall (ES) metric (also known as ‘conditional VaR’ or ‘tail loss’) replaces VaR and stressed VaR. • ES captures the tail risks that are not accounted for in the VaR measures. It calculates the average loss above a certain confidence level. • ES accounts for market illiquidity risk and takes into account the different liquidity horizons of various instruments.	• More risk sensitive than Basel 2.5. • A ‘sensitivity’ is the change in the value of an instrument given a small movement in a risk factor that affects the instrument’s value. For example, banks are required to calculate the change in value of their financial instruments if there was a 1 basis point move in interest rates. • Uses delta risk (potential loss due to a small change in price or rate); vega risk (potential loss due to a change in implied volatility of an option); and curvature risk (the potential incremental loss beyond delta risk when large movements occur in risk factors of instruments that have optionality).
Capital requirement is the sum of the capital requirement for each of three components:	Capital requirement is the sum of the capital requirement for each of three components:
1 an ES metric for modellable risk factors; 2 a non-modellable risk factor requirement for market risk factors which are not suitable for modelling; 3 a default risk capital requirement for credit and equity positions.	1 sensitivities-based methods similar to a stress test; 2 residual risk add-on for any other risk not addressed in 1; 3 a default risk capital requirement with weights assigned that are based on credit rating.

Source: Adapted from Basel Committee on Banking Supervision (2019b).

¹¹ More information on this and subsequent reforms can be found in BCBS (2019a).

summarises the key elements of the internal and standard models approaches under the January 2016 market risk framework.

Bank internal models must also pass two validation tests on an ongoing basis, or else use the standard model. The tests determine if the internal model: (i) measures appropriately the risk that drives the daily profit and loss of the trading desk ('P&L attribution'); and (ii) estimates a risk that is sufficiently conservative to cover observed trading losses ('backtesting').

12.7.1 Value-at-risk (VaR)

VaR is the principal portfolio measure of market risk, and provides an estimate of the potential loss on the current portfolio from adverse market movements. It was originally developed by JPMorgan in the context of their RiskMetrics™ product and builds on modern portfolio theory. VaR is a statistical measure of potential trading revenue volatility, and a change in the general level of VaR would normally be expected to lead to a corresponding change in the volatility of daily trading revenues. The distinguishing feature of VaR is that it uses the volatility of assets.

The basic formula of VaR is as follows:¹²

$$\text{VaR}_x = V_x (dV/dP) \Delta P_t \quad (12.13)$$

where:

V_x = the market value of portfolio x

dV/dP = the sensitivity to market prices movements per £ of market value

ΔP_t = the adverse price movement over a specific time horizon t (under the Basel Agreement, $t = 10$ days).

It expresses the 'maximum' amount a bank might lose, to a certain level of confidence (q), as a result of changes in risk factors (i.e. changes in interest rates, exchange rates, equity and commodity prices). The basic time period t and the confidence level q are the two major parameters that should be chosen in an appropriate way. The time horizon can differ from a few hours for an active trading desk to a year for a pension fund. In simple terms, the VaR approach aims to answer the following question: 'How much can I lose with x per cent probability over a pre-set horizon?'

Suppose that a bank portfolio manager has a daily VaR equal to £1 million at a 99 per cent confidence interval. This means that there is only one chance in 100 that a daily loss bigger than £1 million occurs under normal market conditions.

The calculation of VaR specified in equation (12.13) involves several assumptions:

- prices of financial instruments are assumed to be normally distributed;
- price changes are assumed to be statistically uncorrelated;
- the volatility (standard deviation) of the price or rate changes is stable over time;
- the interrelationship between two different price movements follows a joint normal distribution.

Box 12.6 gives an example of VaR.

¹² A general introduction to VaR can be found in Jorion (2007).

BOX 12.6 EXAMPLE OF VaR

Suppose a portfolio manager manages a portfolio that consists of a single asset. The return of the asset is normally distributed with annual mean return of 15 per cent and annual standard deviation of 25 per cent. The value of the portfolio today is £100 million. We want to answer various simple questions about the end-of-year distribution of the portfolio's value:

- 1 What is the probability of a loss of more than £20 million by year-end – i.e. what is the probability that the end-of-year value is less than £95 million (£115 million minus £20 million)?
- 2 With 1 per cent probability what is the maximum loss at the end of the year? This is the VaR at 1 per cent.

To answer these questions keeping in mind the assumptions made above, we need to employ a statistical package (for example, Microsoft Excel):

- 1 In Excel, you need to employ the formula giving standard normal cumulative distribution. In this example:

$$\text{NORMDIST}(95, 115, 25, \text{TRUE}) = 0.211855$$

The probability of a loss of more than £20 million is 21 per cent.

- 2 In Excel, you need to employ the formula giving the inverse of the normal cumulative distribution. In this example:

$$\text{NORMINV}(0.01, 115, 25) = 56.841325$$

There is 1 per cent probability that the end-of-year value will be less than £56.84 million, which means that the maximum loss is equal to:

$$£115 \text{ million} - £56.84 \text{ million} = £58.16 \text{ million}$$

Most VaR calculations, however, are not concerned with annual value-at-risk. The main regulatory and management concern is with the loss of portfolio value over a much shorter period (typically several days).

After the introduction of market risk measurement in the 1996 Amendment to the Basel I Capital Adequacy Accord, regulators have encouraged the use of VaR. The required VaR measure is for every ten days with a confidence interval of 99 per cent. However, the ten-day VaR measure takes no account of the mitigating action that can be taken in the event of adverse market moves, nor does it express the worst result that could occur as a result of extreme, unusual or unprecedented market conditions. The absolute level of VaR should not, therefore, be interpreted as the likely range of daily trading revenues.

As there is no general consensus as to the 'best way' to carry out a VaR analysis, the approach allows various options for the choice of the underlying frequency distributions:

- parametric methods (e.g. RiskMetricsTM);
- non-parametric methods (historic approach);
- simulation approaches (such as using Monte Carlo techniques);
- hybrid models (e.g. historic simulation with parametric model).

Because of the different approaches that might be followed, i.e. financial institutions may use different confidence levels or holding periods, may have different sources of historical data or use longer or shorter time-series and may use approximate changes in individual risk factors following different distribution, direct comparisons between VaR numbers produced by different institutions can be misleading. A recent formal survey of US and international commercial banks suggests that the most popular VaR estimation methodology is *by far* historical simulation, followed by the Monte Carlo-based method (Perignon and Smith, 2010).

In order to determine the overall or net position, a portfolio can be divided according to its sensitivity to certain risk (the so-called Greeks):

- (a) Delta risk (absolute price risk): the risk that the price of the underlying asset will change.
- (b) Gamma risk (convexity risk): allows for the existence of a non-linear relationship between the change in the price of the underlying asset and the change in the value of the portfolio.
- (c) Vega risk (volatility risk): the risk arising from a change in the expected volatility in the price of the underlying instrument.
- (d) Theta risk (time-decay risk): the risk of a change in the value of the portfolio simply connected with the passing of time.
- (e) Rho risk (discount risk): the risk associated with a change in the risk-free rate.

To arrive at a VaR, the portfolio components are disaggregated according to the above risk factors, netted out and then re-aggregated.

To illustrate how VaR is reported by banks, Table 12.4 reports the figures from UBS's 2019 annual report by business division and corporate centre. It shows that UBS group's average management VaR (1-day, 95 per cent confidence level) decreased to US\$11 million from US\$12 million in the previous year.

Some authors (Taleb, 1997; Daniellson, 2000, 2002) have expressed the following concerns over the use of VaR:

- VaR does not give the precise amount that will be lost.
- The assumption that financial returns are normally distributed and uncorrelated may not hold.

Table 12.4 Value-at-risk at UBS (1-day, 95% confidence, 5 years of historical data)

(US\$ million)	Year ended 31 December 19				Year ended 31 December 18			
	Min	Max	Avg.	31 December	Min	Max	Avg.	31 December
Global Wealth Management	0	1	1	1	0	2	1	1
Personal and Corporate Banking	0	0	0	0	0	0	0	0
Asset Management	0	0	0	0	0	0	0	0
Investment Bank	4	17	9	7	4	25	11	10
Corporate centre	4	8	5	5	4	7	5	6
Diversification effect			(5)	(4)			(5)	(5)
Total management VaR, Group	6	18	11	9	5	26	12	12

Source: UBS (2019).

- VaR measures are seemingly easy to manipulate.
- It does not provide an indication of the probability of a bank failure.
- If all traders are using the same approach to minimise market risk, this can result in increased liquidity risk.

One of the main concerns of prudential regulators is the fact that VaR fails to capture tail risks, hence giving perverse incentives to banks. In fact, the reputation of VaR was seriously damaged in the 2007–2009 financial crises as banks and policy makers realised that markets under stress can generate losses well above the maximum predicted by VaR. The crisis demonstrated that providing insurance against certain tail events was not a ‘risk-less’ strategy as when these tail events occur, they can result in very large unexpected losses.

For these reasons, **backtesting** (that is, an *ex-post* comparison of the risk measure generated by the model against actual daily changes in portfolio value) has been always advocated by the BCBS. Given the limitations of VaR highlighted above, most banks also employ scenario analysis and stress testing. In simple terms, backtesting compares actual revenues arising from closing positions (i.e. excluding intra-day revenues, fees and commissions) with the VaR calculated on these positions, and is used to monitor the quality of the VaR model.

Figure 12.7 shows these daily revenues and the corresponding 1-day 99 per cent confidence level regulatory VaR for UBS over 12 months of 2019.

As illustrated in Figure 12.7, the revenue volatility over 2019 was within the range predicted by the VaR model. If we focus on the exceptions, these occur when backtesting revenues are negative and the absolute value of those revenues is greater than the previous day’s VaR. The figure shows that there were no backtesting exceptions in the second half of 2019. The total number of negative backtesting exceptions within the most recent 250-business-day window decreased from 1 to 0 by the end of the year.

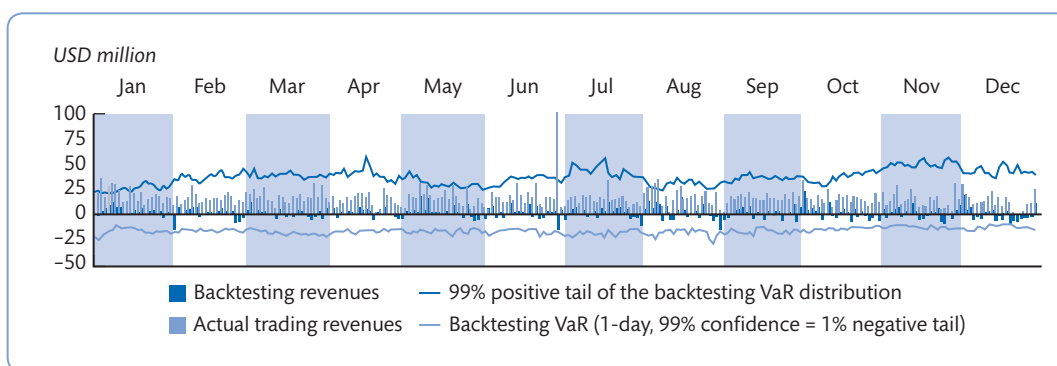


Figure 12.7 Backtesting VaR (UBS), 1 January 2019–31 December 2019

Notes: Excludes non-trading revenues, such as valuation reserves, commissions and fees, and revenues from intraday trading. Includes backtesting revenues from intraday trading. Based on Basel III regulatory VaR, excluded CVA positions and their eligible hedges that are subject to the standalone CVA charge. CVA = Credit Valuation Adjustment.

Source: UBS (2019), p. 141.

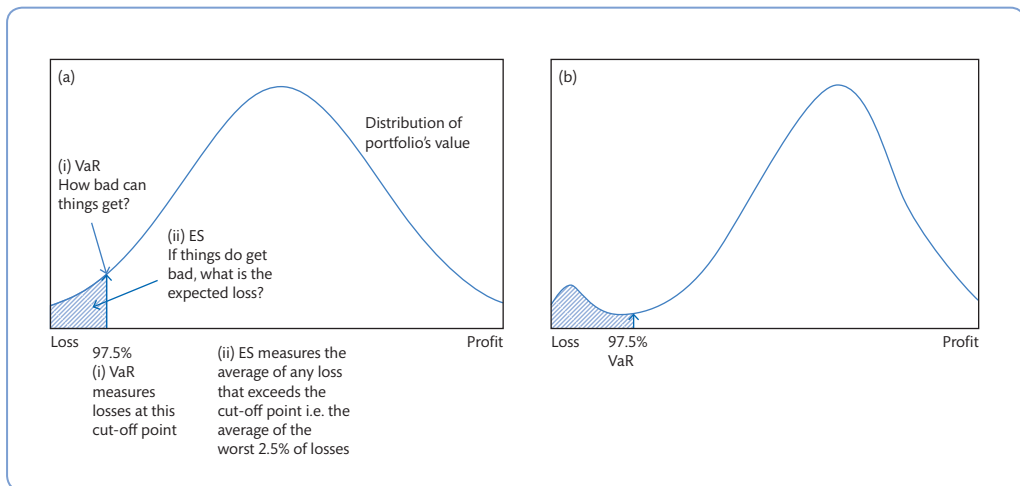


Figure 12.8 Comparison between value-at-risk and expected shortfall

Source: Adapted from Basel Committee on Banking Supervision (2019b) and Risk.net

Scenario analysis and stress-testing practices are based on simulated forecasts of plausible unfavourable scenarios to compute how much a bank would lose in the event of a ‘worst-case scenario’ and tests the bank’s ability to withstand possible shocks. As discussed in the previous section, the revised internal model approach under the January 2016 market risk framework replaces the reliance on VaR with ES. What are the differences between the two? Essentially, the VaR answers the question ‘how bad can things get?’ and represents the maximum risk losses that could potentially be realised at an established level of confidence over a set time horizon (e.g. a 97.5 per cent VaR measures a loss that is expected to be exceeded only 2.5 per cent of time). Conversely, ES answers the question ‘if the things get bad, what is the expected loss?’ Therefore, it calculates the average loss above a certain confidence interval (e.g. a 97.5 per cent ES measures the average of the worst 2.5 per cent of losses). Figure 12.8(a) helps to visualise these differences.

Assume that Figure 12.8(a) and (b) show the probability distribution for the profits and losses of two different portfolios over a certain time period. The two portfolios have the same VaR but the portfolio in Figure 12.8(b) is significantly riskier than the one in Figure 12.8(a). This means that the ES will be much higher for the portfolio in Figure 12.8(b) than the one in Figure 12.8(a).

12.7.2 Risk-adjusted return on capital (RAROC)

The concept of **risk-adjusted return on capital (RAROC)** was first introduced by Bankers Trust in the late 1970s as a planning and performance management tool, in the context of risk-adjusted performance measurement (RAPM). Several approaches, defined as ‘*asset-volatility-based approaches*’ as opposed to the traditional risk-based and ROE-based approaches were developed in response to the need to target shareholders’ value and to allocate banks’ internal resources more efficiently. The RAROC measures the risk inherent in each banking activity, and the risk factor is computed taking into account the asset price volatility, calculated on historical data. An interesting feature of RAROC is that it can be employed to estimate the relative capital allocation of all types of banking risks.

$$\text{RAROC} = \frac{\text{Revenues} - \text{Cost} - \text{Expected losses}}{\text{Total equity capital}} \quad (12.14)$$

In the context of the Basel II Capital Accord, which encouraged the use of internal risk models to set banks' capital requirements, the RAROC measure helped bank managers to assess in what areas they should allocate more capital. However, when Basel III was phased in, considering the major credit losses and banking crisis of the time, it was clear that also the existing approaches to RAROC needed to be revised to take account of **tail risk** and stress testing. Tail risk can be defined as a higher than anticipated risk of an investment moving more than three standard deviations away from the mean. Box 12.7 shows that despite the progress with the vaccine, the Covid-19 pandemic is seen by fund managers as the main tail risk.

BOX 12.7 INVESTORS BET VACCINE-LED RECOVERY WILL LIFT EMERGING MARKETS

Investors are betting emerging markets will be among next year's winning asset classes, wagering a coronavirus vaccine will help spur a global economic recovery, according to a survey of fund managers by Bank of America.

The monthly report, which gathered views of fund managers collectively holding \$526bn in assets, highlighted that the bullishness on emerging markets is part of a broader rotation theme, which has also boosted small-caps, value stocks and banks.

Investors began switching from pandemic winners, such as technology stocks, into sectors tied to a rebound in the economy after Pfizer and BioNTech announced strong early results from the trial of their Covid vaccine on November 9. A similar vaccine from Moderna also proved to be effective in its trial, the company said this week.

Emerging market equities have climbed 2.3 per cent since the Pfizer announcement, taking their gain for the year to just under 8 per cent, according to MSCI.

Bank of America said half of the interviewees indicated emerging markets as their favourite trade for 2021, ahead of the S&P 500 and oil. Meanwhile, a growing number of investors believe emerging market currencies are undervalued.

"The vaccine announcements provided another boost for emerging market risk sentiment," said Andreas Kolbe, a strategist with Barclays. "These results have likely removed some pessimism about the medium-term trajectory of the virus and its consequences on the global economy."

He added lower volatility should boost emerging market assets, particularly currencies and credit. "We expect emerging market local markets to rally across the quality spectrum," he said.

The progress in the fight to curb the spread of the virus has helped to bolster risk appetite. The amount of cash investors are keeping in their portfolios fell to 4.1 per cent, according to the bank, similar to the level in January before coronavirus rocked global financial markets.

Michael Hartnett, chief investment strategist at BofA, said the decline in cash levels was often a signal of investor exuberance.



BOX 12.7 Investors bet vaccine-led recovery will lift emerging markets (*continued*)

“We are starting to see some extreme measures of sentiment, as cash levels are low and equity allocation is high,” he said.

Despite the progress in the fight against the virus, the survey showed 41 per cent of fund managers saw the Covid-19 pandemic as the main “tail risk” although they now expect a “credible vaccine” to be delivered in mid-January, one month ahead of the previous timeline.

Investors polled by the investment bank warned that bets on large US technology companies remained a crowded trade. The pandemic has benefited many of those businesses as consumers live more of their lives from home and use their computers to work or order food. Growth companies such as Amazon, Microsoft and Facebook have posted strong gains since the coronavirus outbreak.

Still, about one in four investors predicted value stocks would outperform growth stocks, the highest level since February 2019 and a sign that they see a brighter outlook for the real economy.



Source: Carnevali, D. (2020) ‘Investors bet vaccine-led recovery will lift emerging markets’, *Financial Times*, 17 November.

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12.8 Managing operational risk

The Basel Committee on Banking Supervision (BCBS, 2011a) broadly defines operational risk as ‘the risk of loss resulting from inadequate or failed internal processes, people and systems or from external events’. In general terms, this is the risk associated with the possible failure of a bank’s systems, controls or other management failure (including human error). This definition focuses on the causes of operational risk, and it is aimed at facilitating operational risk measurement and management.

Various factors have contributed to increase the operational risk in the environment where banks operate over recent years. External forces include the progress in trading technology (e.g. high-frequency trading) and the increased complexities and interconnectedness in the financial marketplace globally. Losses can be quite substantial and recent research has warned that cyber- and IT-related risk are now a main concern for banks.

The increased importance of operational risk in financial institutions was recognised by the Basel Committee and a new capital charge was incorporated in Basel II. Since then, significant progress has been made in setting a framework for an effective measurement and management of operational risk in banking. In August 2020 the Basel Committee published two consultative documents: ‘Revisions to the principles for the sound management of operational risk’ (BCBS, 2020b) and ‘Principles for operational resilience’ (BCBS, 2020c). The former focuses on the importance of adopting an integrated risk management framework for sound operational risk that includes governance, risk management environment, information and communication technology, business continuity planning and the role of disclosure. The latter concentrates on the increased potential for disruptions to the normal banking business from ‘pandemics, natural disasters, destructive cyber security incidents or technology failure’.

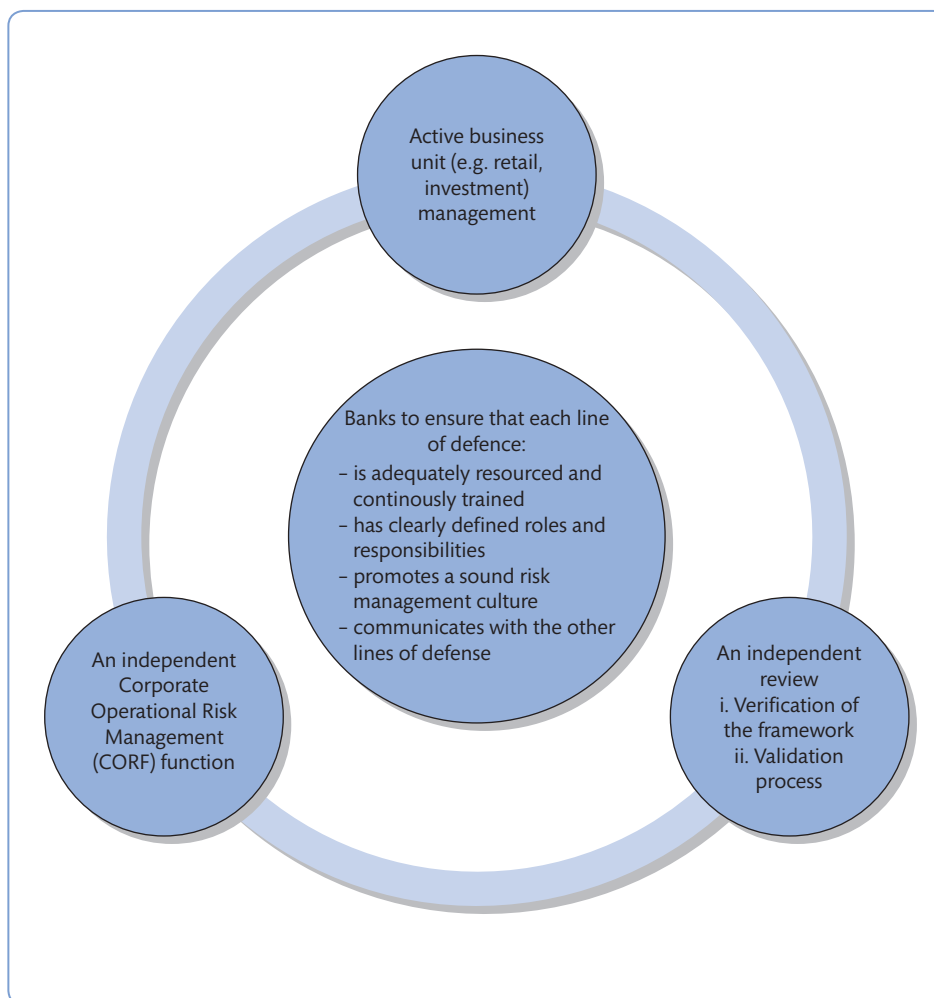


Figure 12.9 The three lines of defence for sound operational risk management

Source: Adapted from Basel Committee on Banking Supervision (2011d, 2020b).

Following these documents, the sound management of operational risk should rely on three lines of defence as illustrated in Figure 12.9.

The first line of defence is business line management, which is responsible for identifying and managing the risks inherent in the products, activities, processes and systems for which it is accountable. The second line of defence is the creation of a specialised and independent internal committee, the corporate operational risk function (CORF), responsible for the design, maintenance and ongoing development of the operational risk framework within the bank. CORF's functions include operational risk measurement and reporting processes, risk committees and responsibility for board reporting. The third line of defence is an independent review of the bank's operational risk management controls, processes and systems. It includes two interrelated aspects: (i) periodic verification and (ii) validation. The former is usually carried out by the bank's internal and/or external audit and is in charge of testing the effectiveness and consistency of the operational risk management framework. The latter is an independent process performed by audit and/or external staff and should confirm the

robustness and reliability of the quantification procedures and risk measurement methodology used by the bank. Banks will have to ensure that the lines of defence can function effectively and, for example, ensure that they are well resourced, appropriately trained and that there is a clear identification of roles and responsibilities.

In line with the management of other banking risks, a capital provision should be set aside to cover for unexpected losses due to operational risk. However, explicit pricing of the losses relating to operational risk is less common.

The Basel II framework proposes three methods for calculating operational risk capital charges, which present increasing sophistication and risk sensitivity:

- 1 the *basic indicator approach*;
- 2 the *standardised approach*;
- 3 the *advanced measurement approach*.

The basic approach allocates capital using a single indicator (gross income) as a proxy for a bank's overall operational risk exposure. Gross income is defined as net interest income plus net non-interest income and the bank is expected to set aside a capital provision to cover for unexpected losses due to operational risk equal to the average of the previous three years of a fixed percentage ('alpha') of positive gross annual income.

In the standardised approach, a bank's activities are divided into eight standard business units and business lines and all use gross income as a common indicator. These are:

- 1 corporate finance (18 per cent);
- 2 trading and sales (18 per cent);
- 3 retail banking (12 per cent);
- 4 commercial banking (15 per cent);
- 5 payment and settlement (18 per cent);
- 6 agency services (15 per cent);
- 7 asset management (12 per cent);
- 8 retail brokerage (12 per cent).

Within each business line, the capital charge is calculated by multiplying a bank's broad financial indicator by a percentage factor 'beta' (in brackets above). This is a rough proxy for the relationship between the industry's operational risk loss experience for a given business line and the broad financial indicator for the bank's activity in that business line. The total capital charge is the sum of the capital charges in each business line.

The advanced measurement approach (AMA) allows individual banks (subject to supervisory approval) to employ their own internal operational risk measurement systems using quantitative and qualitative criteria to calculate the required capital. Banks apply different AMA models depending on the way they combine the following four 'data elements':

- 1 Internal loss data.
- 2 External loss data.
- 3 Scenario analysis.
- 4 Business environment and internal control factors.

The Basel Committee on Banking Supervision (2011e) recognises that there are various ways that an AMA model can be constructed to use the four data elements listed above and

banks should ensure that the resulting level of operational risk capital is adequate given the banks' level of exposure. It also suggests that the most popular methods in modelling operational risk are the loss distribution approach (LDA) followed by scenario-based approaches (SBA).

In terms of actual measurement methodologies, the Basel Committee has identified three sub-approaches that banks can choose from:

- 1 *Scorecard approach* – relies primarily on qualitative analysis across business lines and less on historical loss data in determining capital. The process is constantly updated and aims at creating risk indicators that are combined into a score that reflects the degree of the operational risk. Risk loss data are used to validate results of scorecards.
- 2 *The loss distribution approach* – this is based on the historical database of operational loss for capital calculation. It requires the determination of the probability distribution of the frequency of operational loss events and the conditional probability distribution of the severity of operational losses given an operational loss event. This is calculated for each business unit and type of event; the capital charge is based on the sum of all operational risk for each business line/event type.
- 3 *Internal measurement approach* – see Box 12.8.

BOX 12.8 STRUCTURE OF THE INTERNAL MEASUREMENT APPROACH

In the internal measurement approach, a bank's activities are categorised into a number of business lines, and a broad set of operational loss types are defined and applied across business lines. Within each business line/loss type combination, the supervisor specifies an exposure indicator (EI), which is a proxy for the size (or amount of risk) of each business line's operational risk exposure.

For each business line/loss type combination, banks measure, based on their internal loss data, a parameter representing the probability of loss event (PE) as well as a parameter representing the loss given that event (LGE). The product of EI*PE*LGE is used to calculate the expected loss (EL) for each business line/loss type combination.

The supervisor supplies a factor (the 'gamma term', γ) for each business line/loss type combination, which translates the expected loss (EL) into a capital charge. The overall capital charge for a particular bank is the simple sum of all the resulting products. This can be expressed in the following formula:

$$\text{Required capital} = \sum_i \sum_j [\gamma(i,j) * EI(i,j) * PE(i,j) * LGE(i,j)] \quad (12.15)$$

where:

i = the business line

j = the risk type.

To facilitate the process of supervisory validation, banks supply their supervisors with the individual components of the expected loss calculation (i.e. EI, PE, LGE) instead of just the product EL. Based on this information supervisors calculate EL and then adjust for unexpected loss through the gamma term to achieve the desired soundness standard.

Source: Adapted from Basel Committee on Banking Supervision (2001b), p. 8.

In 2016, the Basel Committee proposed to reorganise the method of assessing operational risk (BCBS, 2016b) by introducing one single risk-sensitive standardised measurement approach (SMA) for all banks to replace all standardised approaches and the AMA. After receiving extensive criticism, the Committee published a final formula in 2017 (BCBS, 2017), but the industry currently lacks clarity and the final implementation was postponed initially to 2022 and then, due to the Covid pandemic, to 2023 (BCBS, 2020d).

12.9 International risk assessment

Banks engaged in international activities face a plethora of risks, in addition to the ones already discussed, including among others foreign currency risk, regulatory risk, strategic and reputation risk. These risks are not mutually exclusive and any product or service provided either domestically or internationally may expose the bank to multiple risks. For banks either contemplating an international investment or those already with substantial overseas operations the risks associated with operating in a foreign country need to be evaluated. Put simply, firms that operate internationally have to evaluate the country risk associated with their investments and/or overseas operations.

Country risk is the risk that economic, social and political conditions and events in a foreign country will adversely affect a firm's commercial/financial interests. In addition to the adverse effect that deteriorating macroeconomic conditions and political and social instability may have on the returns generated from an overseas investment, country risk also includes the possibility of nationalisation or expropriation of assets, governments revoking licences, imposition of exchange controls, and the likelihood of currency depreciation or devaluation.

Country risk can have a critical effect on a firm's international activities and therefore needs to be explicitly taken into account in the risk assessment of all overseas investments/activities. Even the risk associated with what are perceived to be the most stable (or safe) investments will increase if, for instance, the political or macroeconomic conditions change and cause the exchange rate to depreciate resulting in lower cash flows (and therefore profits) from overseas investments. Country risk is also not necessarily limited to a firm's exposure to overseas operations. A firm may have commercial relationships in its home country with a foreign firm that may be subject to such risks. For instance, country risk factors should also be taken into account, when assessing the creditworthiness of domestic trade creditors. One should also be aware that country risk factors are critically important for all types of international firms, non-financial and financial. Borrowers in higher-risk countries pay higher premiums for their debt compared with those located in lower-risk countries.

In banking, country risk is regarded as the exposure to a loss in cross-border lending, caused by events in a particular country that are (at least to some extent) under the control of the government but are not under the control of a private enterprise or individual. This contrasts with what is known as sovereign risk, which relates to the risk associated with a government default on bond or loan repayments. A broader definition of country risk relates to any loss associated with international activity due to adverse changes in the overseas operating environment beyond the control of the firm. Transfer risk is another form of risk that is believed to be one of the most important drivers of country risk. This is simply the risk associated with the restriction of foreign payments from overseas to the home company or bank. Transfer risk refers to restrictions on payments between private agents whereas

sovereign risk is associated with a government default on payments. In reality, sovereign and transfer risks are closely related as a government default on payments may lead private parties to renege on their payment obligations – especially if the government default leads to a major depreciation or crisis scenario.

12.9.1 Managing country risk

In order to effectively control the level of risk associated with their international operations, firms must have in place a procedure that systematically evaluates the country risk features of its business. This includes having in place a country risk evaluation process that has:

- effective oversight by senior managers;
- appropriate risk management policies and procedures;
- an accurate system for reporting change in country risk and potential exposures;
- an effective process for analysing country risk;
- a country risk-rating system;
- regular monitoring of country conditions.

While the details and complexity of country risk assessment will vary from bank to bank, senior management must be suitably qualified to evaluate the bank's international activities.

If country risk is to be managed effectively then senior bank managers, up to board level, must oversee the process. It is likely that a team of senior project appraisers will review the bank's international operations in order to ensure that they are consistent with the company's major strategic objectives. Decisions to extend international operations and exposure to different countries' risk will ultimately be a decision for the company board, as they should have a view of the sorts of country exposure required and it is up to the board to make sure that country risk (as well as other risks) is effectively managed.

It is the responsibility of senior bank management to implement policies and procedures for managing country risk. This involves:

- identifying investments and other activities exposed to country risk;
- identifying desirable and undesirable opportunities that can be used to complement or be substituted for current operations resulting in a reduction of country risk;
- establishing country risk limits if necessary;
- identifying clear lines of responsibility and accountability for country risk management decisions.

Senior management are ultimately responsible for country risk management policies, standards and practices and also need to make sure that these are communicated effectively to relevant parts of the organisation.

In order to effectively manage country risk, banks need to have reliable systems for capturing and categorising the volume and nature of their foreign activities. Such a reporting system should cover all aspects of the company's international operations. Banks, for example, have to have country exposure reporting systems to support regulatory reporting of foreign exposures requirements.

The level of resources dedicated to the country risk analysis process will vary from bank to bank, depending on the size and sophistication of the company's international activities.

In order to construct an effective country risk evaluation process senior managers need to ask the following questions:

- Is there a quantitative and qualitative assessment of the risk associated with each country in which the firm is conducting or planning to undertake activities?
- Is any formal country risk analysis undertaken on a regular basis and are changes in country risk monitored in any way?
- Is the country risk analysis adequately documented, with the findings communicated to the relevant parties?
- Are adequate resources devoted to country risk evaluation procedures?
- Do the company's country risk assessments concur with the risk ratings of third-party assessors, such as rating agencies?

If the answers to these questions are in the affirmative then the bank is well placed to use the results of its country risk analysis effectively in strategic and operational decision making. In order to arrive at a conclusion about the level of country risk faced by a bank, managers need to evaluate the current (and possible future) economic, political and social characteristics of a country. For this they are likely to use some form of country risk-rating system.

12.9.2 Country risk rating

Country risk ratings simply summarise the main findings of the country risk analysis process. While large firms and banks are likely to have teams evaluating country risk, smaller firms are more likely to rely heavily on the country risk assessments done by specialist third-party firms. Because there is a wide range of factors that can affect country risk, it is often difficult for smaller firms to dedicate the relevant resources to assess somewhat complex issues. Macroeconomic and political environments can change rapidly and it is often difficult to keep abreast of these developments, especially if one is considering or monitoring projects in a number of countries. Box 12.9 provides a broad indication of the various factors that affect country risk.

One can see from Box 12.9 that there is a whole host of factors that affect a country's risk rating including various economic, financial and socio-political risks, as well as those risks that may be relevant to the specific bank or firm in question. In quantifying the broad economic/financial and socio-political risks, companies can do their own risk evaluation but can also cross-check these with a variety of ratings calculated by third-party firms. There are many firms that provide services that measure country risk and among the main providers there are also private companies that sell their country risk ratings via the web. Examples include International Country Risk Guide (ICRG), OECD, Fitch, Moody's Investor Services and Standard & Poor's Rating Group. Apart from the OECD, all act as 'rating agencies' and sell their country risk ratings via the web or through other media.

Each of these firms produce risk ratings using a variety of qualitative and quantitative information so as to construct a single index or country risk-rating schedule. For example, International Country Risk Guide (ICRG) compiles monthly data on a variety of political, financial and economic risk factors to calculate risk indices in each of these categories as well as a composite risk index. Five financial, thirteen political and six economic factors are used. Each factor is assigned a numerical rating within a specified range. In the case of ICRG country risk weightings, political risk assessment scores are based on subjective staff analysis